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BIRZEIT UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE DEPARTMENT
SOFTWARE ENGINEERING – COMP433
G#2 FINAL (COMPLETE PROJECT) REPORT

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Team members

Layan Salem: Requirement Engineer

Shahd Loai: Technical Architect

Hala Sabobeh: project manager

Nour Omarii: Programmer

Yara Qaraqe: Secretary

1 Chapter 1: Project Planning and Management

1.1 Names of Editors/writers of the Report

writers: Shahd Loai 1211019, Layan Salem 1221026, Hala Sabobeh 1220322

Editors: Nour Omarii 1221212, Yara Qaraqe 1220505

1.2 Business Title

Hand Watches Shop

1.3 Group Name

G2

1.4 Name of students

- Hala Sabobeh 1220322
- Shahd Loai 1211019
- Nour Omarii 1221212
- Yara Qaraqe 1220505
- Layan Salem 1221026

1.5 Role of each member

- Hala Sabobeh (Project manager)
- Shahd Loai(Technical Architect)
- Nour Omarii (Programmer)
- Yara Qaraqe (Secretary)
- Layan Salem (Requirement Engineer)

1.6 Project management strategy

Meetings: Our plan was to meet at least once per week. As team members reside in different cities and have differing college break schedules, we found it most efficient to conduct our meetings online. We primarily used Google Meet for our weekly coordination sessions. To maximize productivity during available breaks, we also organized intensive work periods, such as a dedicated week during the winter break where we held daily meetings of approximately 3 hours each.

Decisions: We employed a collaborative and democratic decision-making process. Every significant decision involved gathering all team members' opinions, discussing each one thoroughly, and then voting to select the option deemed best for the project. In the event of a tie, the Project Manager held the final deciding vote to ensure we could move forward decisively.

Process Models: We adopted a hybrid approach, utilizing both Waterfall and Agile methodologies. This allowed us to benefit from the structured phases and clear milestones of Waterfall while incorporating the flexibility, iterative feedback, and adaptability of Agile to ensure the project ran smoothly and could respond to challenges effectively.

1.7 Project manager report

This was a significant project for our team, and the responsibilities were divided according to each member's role, ensuring a balanced workload and comprehensive coverage of all software engineering phases. We collaborated closely on group tasks and utilized tools like Lucidchart for real-time diagramming. The following summarizes the contributions of each team member:

• **Hala Sabobeh (1220322) – Project Lead:** As the Group Manager, my responsibilities encompassed establishing project milestones, coordinating and scheduling meetings, and fostering clear communication among team members. I led the creation of the Business Description, User Requirements Specifications, Use-Case Activity Modelling, and System and Component Modelling. I also facilitated discussions on System Requirements Specifications and reviewed and edited the Actor Analysis, Use-Case Modelling, System Class Modelling and Analysis, System Design Goals, System and Architectural Design, and System and Deployment Modelling. For the core system analysis, I authored the "Home Delivery with Tracking" scenario and its corresponding "Update Order Status" Use-Case Specification and Activity Model. I also led the development of the overall Use Case Model for the system.

• **Layan Salem (1221026) – Requirements Engineer:** As the Requirements Engineer, she was responsible for defining user and system requirements through structured analysis. Layan led the Actor Analysis, defining all primary and secondary actors in the system. She authored the "Browse, Search, and Filter Watches" scenario and developed the corresponding "Browse Products" Use-Case Specification and Activity Model. She actively participated in discussions on Use-Case Modelling and Effort Estimation, and reviewed the System Class Modelling and Activity Modelling deliverables to ensure alignment with user needs.

• **Shahd Loai (1211019) – Technical Architect:**As the Technical Architect, she spearheaded the formulation of Design Goals and provided justification for architectural decisions. Shahd authored the "Request After-Sales Service" scenario, focusing on system state and business logic, and created its detailed Use-Case Specification. She contributed to the Use-Case Modelling and Actor Analysis through review and editing, and led critical discussions on System Class Modelling and Analysis to ensure a robust design foundation.

• **Nour Omarii (1221212) – Programmer:** As the Programmer, she was responsible for analyzing system processes and interactions, particularly from a backend and operational perspective. Nour authored the "In-Store Watch Purchase (Staff Perspective)" scenario and the "Update Inventory" Use-Case Specification. She led the Activity Modelling task, coordinating the team's effort in creating detailed process flows. She also participated in discussions on Actor Analysis and Use-Case Modelling, and reviewed Architectural and Component Modelling deliverables.

• **Yara Qaraqe (1220505) – Secretary:** As the Secretary, she played a vital role in documenting core user workflows and ensuring clarity in process definition. Yara authored the primary "Online Watch Purchase" scenario, including its normal, alternative, and error flows, and developed the "Complete Checkout and Payment" Use-Case Specification and Activity Model. She supported the Project Lead in organizing deliverables and actively participated in discussions for Actor Analysis, Use-Case Modelling, and Activity Modelling.

As fourth-year Computer Engineering students, this project was particularly challenging and insightful. It required significant effort to synchronize our work and find common meeting times. However, it provided an excellent opportunity to experience the end-to-end process of software specification and design. It significantly enhanced our technical documentation, teamwork, and systems analysis skills, giving us a practical understanding of how complex requirements are translated into structured models.

1.8 Contribution by each group member to the project

- Lead: Hala Sabobeh (1220322)

The project was completed successfully through good leadership, organization, and teamwork. As the team lead, I was responsible for organizing tasks, following up on progress, and making sure deadlines were met. I also helped in coordinating communication between team members and reviewing the final work to ensure quality and consistency. This role helped me improve my leadership and teamwork skills.

- Technical Architect: Shahd Loai (1211019)

As the Technical Architect, I worked on designing the system architecture and making sure all technical parts were consistent. I contributed to user and system requirements, effort and time estimation, actor analysis, and use-case modelling. I also led the system class modelling and architectural design, and worked on sequence diagrams and scenarios. This role helped me gain strong technical and design experience.

- Programmer: Nour Omarii (1221212)

As the Programmer, I contributed to implementing system-related logic and supporting the technical development of the project. I participated in reviewing system requirements and use-case models to ensure they were technically feasible. I also helped in developing and reviewing activity and sequence diagrams and supported system class and component modelling to ensure correct and clear implementation.

- Secretary: Yara Qaraqe (1220505)

As the Secretary, I was responsible for documenting meetings, organizing files, and keeping clear communication between team members. I recorded decisions, task assignments, and updates, and helped organize the project report in a clear and structured way.

- Requirement Engineer: Layan Salem (1221026)

As the Requirements Engineer, I worked on defining and documenting the system requirements. I led the effort and time estimation task and contributed to system design goals. I also worked on scenario analysis, use-case specifications, activity modelling, and sequence modelling to ensure the system was clearly defined and well-structured.

2 Chapter 2: Requirement Elicitation, Analysis and Modelling

2.1 Requirement statement/Business description.

Description

The Hand Watches Shop is an established online business, specializing in the sale of wristwatches from well-known brands for men, women, and children. All inventory is managed and dispatched from our secure storage facilities. Customers can browse, purchase, and arrange delivery entirely online, with payment options including credit card and cash-on-delivery. In addition to sales, we provide mail-in basic maintenance services such as battery replacement and strap adjustments, alongside comprehensive after-sales support for returns, warranties, and customer inquiries. We cater to individuals seeking reliable watches and convenient maintenance, delivering directly to their door with a focus on trust and service.

Business Services

1. Selling Wristwatches

- Offers wristwatch models categorized for men, women, and children.
- Customers can compare and select watches based on brand, model, price, color, and features.
- Sales are conducted in-store with assistance from store staff.

2. Maintenance and Repair Service

- Provides battery replacement, strap adjustments, and simple repairs for watches purchased from the store.
- Maintenance services are handled by a trained watch technician using the store's tools.

3. Delivery Service

- Prepares and packs watches for home delivery.
- Coordinates with a delivery company for delivering orders, typically within 1–3 days depending on the city.

4. Payment Service

- Accepts payments in cash or by credit card.
- Payments are made in-store for direct purchases or during order confirmation for delivery orders.

5. After-Sales Support

- Handles returns, warranty claims, defective products, and customer inquiries.
- Ensures customers receive quality assistance post-purchase.

Business Capacities

- Serves 150 customers per month.
- Processes 120 purchase orders per month.
- Can handle 6 customers concurrently during peak hours.
- Offers a selection of exactly 48 watch models in three categories.
- Operates 6 days a week, 9:00 AM to 8:00 PM (closed on Fridays).
- Delivery service is available Sunday to Thursday.

Working Process

Watch Selling Process

1. Customers visit the shop and browse available watches.
2. Compare watches based on brand, price, color, and features.
3. Select a watch and pay via cash or credit card.

Maintenance Process

1. Customer brings the watch to the shop.
2. Technician inspects the watch.
3. Performs required maintenance (battery replacement, strap adjustment, repair).
4. Returns the watch to the customer.

Delivery Process

1. Customer provides order and delivery details.
2. Shop prepares, packs, and checks the watch.
3. Delivery company delivers the watch to the customer within 1–3 days.

Payment Process

Delivery order payments are confirmed at order placement via cash or card.

After-Sales Process

1. Customer submits a return, warranty, or maintenance request.
2. Shop verifies the purchase.
3. Issue resolved through repair, replacement, or refund.

2.2 USER and SYSTEM requirements.

[Group Task] Lead: [Hala Sabobeh]; contributors: [Yara Qaraqe: reviewing], [Layan Salem: discussion], [Shahd Loai: Contact with customers], [Nour Omarii: discussion]

USER REQUIREMENTS

MODIFIED

- **UR1:** The customer shall be able to browse the available watch inventory.
- **UR2:** The customer shall be able to search and filter watches.
- **UR3:** The customer shall be able to purchase a watch.
- **UR4:** The customer shall be able to request after-sales services.
- **UR5:** The customer shall be able to purchase a watch in person at the store.
- **UR6**: The customer shall be able to have their purchased watch delivered to their home.
- **UR7:** The customer shall be able to view feedback and reviews from previous buyers for a specific watch model.
- **UR8:** The customer should view the shop's opening hours and delivery service availability.

[Group Task] Lead: [Layan Salem]; contributors: [Yara Qaraqe: reviewing], [Hala Sabobeh: discussion], [Shahd Loai: Contact with customers], [Nour Omarii: discussion]

SYSTEM REQUIREMENTS

MODIFIED all notes about system requirements

UR1: The customer shall be able to browse the available watch inventory.

- SR1.1: The system shall maintain and present a complete catalog of available watches, including their descriptions, pricing, and stock status.
- SR1.2: The system shall present product information in a clear and organized manner for customer review.
- SR1.3: The system shall provide the ability to order catalog items by attributes including price, release date, and alphabetical order.
- SR1.4: The system shall ensure catalog information reflects current inventory status and updates accordingly.

UR2: The customer shall be able to search and filter watches.

- SR2.1: The system shall support searching the product catalog using keywords.
- SR2.2: The system shall allow filtering of products by price range.
- SR2.3: The system shall allow filtering of products by brand.
- SR2.4: The system shall allow filtering of products by gender classification.
- SR2.5: The system shall allow filtering of products by style or functional category.

- SR2.6: The system shall support the concurrent application of multiple filter parameters.
- SR2.7: The system shall provide the ability to remove all active filter selections.
- SR2.8: The system shall handle scenarios where no products match the search or filter criteria by informing the user.
- SR2.9: The system shall update the displayed product results in response to changes in search or filter criteria.

UR3: The customer shall be able to purchase a watch.

- SR3.1: The system shall enable users to add items in the shopping cart.
- SR3.2: The system shall save cart contents (7 days for registered users, 24 hours for guests).
- SR3.3: The system shall collect and validate customer-provided information, including contact and delivery details, before order confirmation.
- SR3.4: The system shall support cash on delivery, card payments, and digital wallets.
- SR3.5: The system shall deduct purchased items from inventory after payment.
- SR3.6: The system shall present a comprehensive order summary, including all charges and applied discounts, and generate a unique order number upon successful order placement.

UR4: The customer shall be able to request after-sales services

- SR4.1: The system shall support the submission of after-sales service requests
- SR4.2: The system shall select of service type (repair, battery, return, exchange).
- SR4.3: The system shall verify the order number and eligibility (warranty/return period).
- SR4.4: The system shall schedule service dates to customers if needed.
- SR4.5: The system shall provide estimated service completion times.
- SR4.6: The system shall maintain a complete record of all service requests and actions performed for each product.

UR5: The customer shall be able to purchase a watch in person at the store.

- SR5.1: The system shall support in-store sales transactions, requiring authenticated staff access to perform them.
- SR5.2: The system shall support in-store sales using a POS (Point of Sale Interface) and
- require staff login to access POS functions.
- SR5.3: The system shall update inventory immediately after an in-store purchase.
- SR5.4: The system shall register warranty details at purchase.
- SR5.5: The system shall support refunds, cancellations, and exchanges.

UR6: The customer shall be able to have their purchased watch delivered to their home.

- SR6.1: The system shall verify whether the entered delivery address is within the supported delivery area.
- SR6.2: The system shall maintain the validated delivery location for each order.
- SR6.3: The system shall calculate and present an estimated delivery date or time window for the order prior to final confirmation.
- SR6.4: The system shall record the delivery status of each order.
- SR6.5: The system shall make the current status of an order's delivery available to the customer.
- SR6.6: The system shall manage scenarios where delivery is unsuccessful by supporting rescheduling for redelivery or alternate pickup arrangements.

UR7: The customer shall be able to view feedback and reviews from previous buyers for a specific watch model.

- SR7.1: The system shall link all reviews and ratings to the specific purchased product.
- SR7.2: The system shall require a confirmed purchase to allow a customer to submit a review.
- SR7.3: The system shall collect a numerical rating as part of the review submission.
- SR7.4: The system shall collect written feedback as part of the review submission.
- SR7.5: The system shall calculate and present the average rating for each product.
- SR7.6: The system shall present submitted and approved reviews for a product.
- SR7.7: The system shall organize displayed reviews by criteria such as rating or date.

UR8: The customer should view the shop's opening hours and delivery service availability.

- SR8.1: The system shall maintain information regarding the shop's business hours and planned closures.
- SR8.2: The system shall indicate whether the shop is currently open or closed for business.
- SR8.3: The system shall present information regarding the availability of delivery services.
- SR8.4: The system shall present the shop's contact details.
- SR8.5: The system shall present the shop's physical location.
- SR8.6: The system shall allow authorized personnel to update the shop's information.
- SR8.7: The system shall inform customers of any unexpected or temporary closures.
- SR8.8: The system shall present answers to common questions about shop services.

2.3 SCENARIOs

Scenario 1 – Yara Qaraqe

Actor: Customer

Service: Online Watch Purchase

Initial Assumption

Costume browses a watch store website, filters and selects a watch, adds it to her cart, checks out with cash on delivery, and receives an order confirmation.

Normal Scenario

The customer reviews the watches in the shopping cart, including the selected quantities, individual prices, applied discounts, and the total payable amount, and verifies the watch specifications such as brand, model, color, and size. The customer then clicks the "Proceed to Checkout" button, after which the system displays a checkout form requesting the customer's full name, email address, mobile phone number, and delivery address. Once the customer submits the form, the system validates all entered information and checks whether the customer is a registered or returning customer in order to automatically apply any eligible loyalty discounts. The customer selects home delivery as the delivery option, and the system displays a final order summary that includes the product price, delivery fees, applied discounts, and the updated total amount. Instead of choosing online payment, the customer selects Cash on Delivery as the payment method, and the system confirms that payment will be collected upon delivery without requiring card details. The system then displays an order confirmation message, generates a unique order number, reserves the ordered watch by updating the inventory, and sends an email confirmation containing the order details and cash-on-delivery instructions to the customer.

First Alternative

The customer reviews the watches in the shopping cart, including the selected quantities, individual prices, applied discounts, and the total payable amount, and verifies the watch specifications such as brand, model, color, and size. The customer then clicks the "Proceed to Checkout" button, after which the system displays a checkout form requesting the customer's full name, email address, mobile phone number, and delivery address. Once the customer submits the form, the system validates all entered information and checks whether the customer is a registered or returning customer in order to automatically apply any eligible loyalty discounts. The customer selects home delivery as the delivery option, and the system displays a final order summary that includes the product price, delivery fees, applied discounts, and the updated total amount. The customer selects Credit Card as the payment method and is redirected to a secure payment page, where valid card

details are entered and confirmed. The system processes the payment successfully, displays a confirmation message indicating that the order has been placed, generates a unique order number, updates the inventory by deducting the purchased watch quantity, and finally sends an email receipt containing the complete order details to the customer.

Second Alternative

The customer reviews the watches in the shopping cart, including the selected quantities, individual prices, applied discounts, and the total payable amount, and verifies the watch specifications such as brand, model, color, and size. While proceeding to checkout, the customer decides to modify the cart by adding, removing, or changing the quantity of one or more watches. The system updates the shopping cart in real-time, recalculates the total price including any discounts, and ensures inventory availability for the updated items. The customer then clicks the "Proceed to Checkout" button, where the system displays a checkout form requesting personal and delivery information including full name, email address, mobile phone number, and delivery address. Upon form submission, the system validates all entered information and checks for any eligible loyalty discounts, applying them automatically if applicable. The customer selects the preferred delivery option, and the system displays a final order summary reflecting the updated cart contents, delivery fees, discounts, and total payable amount. The customer then selects the preferred payment method (credit card, digital wallet, or cash on delivery), and the system processes the order accordingly. After successful payment or confirmation for COD, the system generates a unique order number, updates inventory based on the modified cart, displays a confirmation message, and sends an email receipt containing the final order details to the customer.

Error Scenario1 - Empty Shopping Cart:

The customer attempts to proceed to checkout without adding any watches to the shopping cart. When the "Proceed to Checkout" button is clicked, the system detects that the cart is empty and prevents checkout. An error message is displayed, informing the customer that at least one item must be added to the cart before proceeding. The system remains on the shopping cart page and highlights the empty state to guide the customer in adding items.

Error Scenario2- Invalid Personal Information:

The customer fills in the checkout form with incorrect or incomplete personal information, such as an invalid email address (e.g., missing "@") or leaving the mobile number field empty. Upon submission, the system validates all fields, detects the errors, and displays specific error messages

next to each invalid or missing field. The customer is prompted to correct the information before proceeding to payment, ensuring that delivery and communication details are accurate.

Error Scenario3- Payment Failure:

The customer selects credit card as the payment method and enters card details that are expired, invalid, or incorrect. When attempting to process the payment, the system communicates with the payment gateway and receives a payment rejection. An error message is displayed, explaining that the payment failed and providing options to retry, use a different card, or select another payment method such as cash on delivery. The system does not generate an order number until the payment is successfully completed.

Error Scenario4- Out-of-Stock Item During Checkout:

While the customer is completing the checkout process, the selected watch becomes out of stock due to simultaneous purchases by other customers. Upon form submission, the system checks inventory, detects insufficient stock, and prevents completion of the order for that item. An error message is displayed, showing the updated stock availability, and the customer is prompted to adjust the quantity or remove the out-of-stock item before proceeding.

Error Scenario5- Delivery Address Outside Coverage Area:

The customer enters a valid personal and delivery address, but it falls outside the store's supported delivery region. Upon submission, the system verifies the address and detects that delivery is not available for that location. An error message is displayed, informing the customer of the delivery restriction, and the checkout cannot proceed until a valid address within the coverage area is entered. The customer may either update the address or choose an in-store pickup option if available.

Scenario 2- Layan Salem

Actor: Customer

Service: Browse, Search, and Filter Watches

Initial Assumption

The Hand Watches Shop system is operational and accessible through web and mobile platforms. The watch inventory database contains up-to-date product information and current stock levels. The customer, whether a guest or a registered user, is on the main catalog page and wants to explore available watches before making a purchase decision. No authentication is required to use the browsing, searching, or filtering service.

Normal Scenario (Successful Outcome)

The customer opens the Hand Watches Shop catalog page, where watches are displayed in a grid format showing thumbnail images, brand names, model names, prices, style categories, and stock status. The customer enters a search term in the search bar, such as a brand name or watch feature. The system searches the inventory database by matching the search term against watch attributes including brand, model, description, and features, then displays a list of matching watches along with the number of results found.

The customer reviews the filtering options provided on the page, which include a price range slider with minimum and maximum values, brand selection through checkboxes, gender selection using radio buttons, style selection through checkboxes, feature selection such as water resistance or chronograph functionality, and a stock availability option to show only in-stock watches. The customer selects multiple filter options according to their preferences. The system processes all selected filters together using combined conditions and updates the displayed results in real time, showing only watches that satisfy all chosen criteria simultaneously.

The customer then selects a sorting option, such as ordering results by price from low to high or by customer rating. The system reorders the filtered results while preserving all active filters. After reviewing the refined list, the customer selects a specific watch. The system retrieves the complete product record and displays the detailed product page, including high-resolution images, technical specifications, full description, price and discounts, real-time stock availability, customer reviews and ratings, estimated delivery time, and recommendations for similar watches. The customer reviews the information and may add the watch to the cart, save it to a wishlist, or return to the results list to compare other watches.

Alternative Scenario (Successful Outcome)

Instead of using the search function, the customer chooses to browse visually. The customer applies filters directly to the full catalog by selecting options such as gender, style, and price range. The system applies these filters to the entire inventory and displays all watches that match the selected criteria. The customer browses the filtered results and selects a watch to view its details successfully.

Error Scenarios:

Error Scenario 1 Unsuccessful Outcome – System Failure:

The customer enters a search term and applies filters. When the system attempts to retrieve matching watches, a database connection failure occurs due to a network or server issue. The system detects the failure and displays an error message stating: "Unable to retrieve watch data at this time. Please try again later." A retry option is provided. If the customer selects retry and the connection is restored, the system displays the results normally; otherwise, the customer must exit or try again later.

Error Scenario 2 Unsuccessful Outcome – Business Constraint:

The customer applies a very narrow combination of search and filter criteria. The system successfully processes the request but finds no matching watches. The system displays a message stating: "No watches found matching your criteria." The system suggests removing or adjusting filters and provides options such as "Remove Last Filter" or "Clear All Filters." The customer modifies the criteria to continue browsing**.**

Error Scenario 3 Unsuccessful Outcome – Inventory Change:

A watch appears in the search results as available. While the customer is reviewing options, another customer purchases the remaining stock of that watch. When the customer selects the watch, the system retrieves the updated inventory record and displays a message stating: "This watch is currently out of stock." The system disables the "Add to Cart" option and offers alternatives such as viewing similar watches or adding the item to a Wishlist. The customer selects an alternative watch or continues browsing.

Scenario 3- Shahd Loai

Actor: Customer

Service: Request After-Sales Service (Repair / Battery Replacement / Return)

Normal Scenario

The customer opens the After-Sales Service page on the website. The system displays a service request form asking for the order number and service details, including a dropdown menu for service type and an optional text field for problem description. The customer enters the order number, and the system retrieves and shows the purchased watch information, such as the model, purchase date, warranty status, and any previous service history.

After verifying the watch information, the customer selects the type of service needed, such as repair, battery replacement, or return. The system automatically checks whether the watch is still under warranty or within the allowed return period. If the service is repair or battery replacement, the system indicates whether the service is free under warranty or if additional fees apply. The customer may add a short description of the problem, such as "Watch not keeping accurate time" or "Battery draining quickly."

Once the customer submits the request, the system validates all entered information. If everything is correct, the system saves the service request, generates a unique service request number, and displays a confirmation message to the customer, such as: "Your service request has been successfully submitted. You will receive an email confirmation shortly."

Alternative Scenario:

In-Person Request: If the customer prefers to request the service in person, they can visit the store and provide the order number to the staff. The staff enters the service request information into the system on behalf of the customer. Once submitted, the system displays a confirmation message similar to the online request.

Return Option: If the customer chooses the return option, the system verifies the return period. If the watch is eligible for return, the system displays clear return instructions, such as packing requirements, return location, and estimated processing time, before completing the request.

Error Scenario1- Invalid Order Number:

If the customer types an incorrect order number or one that does not exist in the system, the system displays a clear error message: "The order number you entered is invalid. Please check the number and try again." The customer cannot proceed with the service request until a valid order number is entered. The system may also provide a link to a page where the customer can check their order history.

Error Scenario2- Expired Warranty or Return Period:

If the customer requests a service such as repair, battery replacement, or return, but the watch is out of warranty or the return period has expired, the system displays a message: "The requested service is not available. The return period for this product has expired or the warranty has ended." The system may suggest alternative options, such as paid repair, battery replacement at a service center, or visiting a physical service location. The customer must choose one of the available options to continue.

Error Scenario3- Missing Required Information:

If the customer does not fill in all the required fields, such as service type or problem description, the system highlights the empty fields and shows a message: "Please complete all required fields before submitting your request." The request cannot be submitted until all required fields are completed. This ensures that the system has enough information to process the service efficiently.

Scenario 4 - Nour Omari

Main Actor: Staff (Employee)

Other Involved Actors: Payment Customer System

Service: In-Store Watch Purchase (Staff Perspective)

Initial Assumption

This scenario assumes that the staff member is already logged into the Point of Sale (POS) system inside the watch store and that the system is functioning correctly. The selected watch is available in the inventory, and the customer is physically present in the store and intends to complete an in-store purchase.

Normal Scenario

The customer enters the watch store and selects a watch they wish to purchase. The staff member opens the POS system and searches for the selected watch or scans it to retrieve its details. The system displays the watch information, including price and availability, allowing the staff member to confirm that the item is in stock. The staff then asks the customer for their preferred payment method, and the customer chooses to pay in cash. The staff confirms the payment in the system, after which the system successfully records the sale, updates the inventory, and registers the warranty information. Finally, the system generates a purchase receipt, and the staff hands the watch along with the receipt to the customer, completing the purchase process.

Alternative Scenario – Card Payment

During the in-store purchase process, the customer may choose to pay using a bank card instead of cash. In this case, the staff enters the payment amount into the POS system, and the system sends the payment request to the bank system for authorization. Once the payment is approved, the system completes the transaction, updates the inventory, registers the warranty information, and generates the receipt. The purchase process is completed successfully using an alternative payment method.

Alternative Scenario – Customer Provides Optional Information

During the in-store purchase process, the staff may ask the customer if they would like to provide personal information such as name and phone number for warranty or future reference purposes. If the customer agrees, the staff enters the information into the POS system. The system saves the customer details and continues the purchase process normally until the transaction is completed and the receipt is issued.

Error Scenario 1 – Watch Not Available in Inventory

During the in-store purchase process, the system may indicate that the selected watch is no longer available in the inventory due to recent sales or delayed inventory updates. When the staff attempts to retrieve the watch details, the POS system displays a message indicating that the item is out of stock. In this case, the purchase process cannot proceed, no transaction is recorded, and the inventory remains unchanged. The staff informs the customer, who may choose another available watch or cancel the purchase.

Error Scenario 2 – Payment Authorization Failure

If the customer chooses to pay by card and the bank system rejects the payment due to authorization failure or insufficient balance, the POS system displays an error message indicating that the payment could not be completed. The transaction is not recorded, and the inventory remains unchanged. The staff informs the customer, who may retry the payment using another card, choose a different payment method such as cash, or cancel the purchase.

Error Scenario 3 – Inventory Retrieval Failure

While the staff is searching for or scanning the selected watch, the system may fail to retrieve the watch information from the inventory database due to a system or database error. The POS system displays an error message indicating that the item information cannot be retrieved at the moment. In this case, the purchase process is paused, no transaction is recorded, and the inventory data remains unchanged until the issue is resolved or the process is restarted.

System State on Completion

If the purchase is completed successfully, the system records the sale, updates the inventory, registers the warranty, and issues a receipt. If the purchase fails or is cancelled due to an error, no sale is recorded and the inventory remains unchanged.

Scenario 5 - Hala Sabobeh

Actor: Delivery Company.

Service: Home Delivery with Tracking.

Initial Assumption

A delivery partner is authenticated and logged into the delivery management system via the website. The system has already assigned them a set of deliveries for the day, each with a confirmed address within the service area, a scanned package barcode, and a scheduled time window.

Normal Scenario

The delivery partner begins by logging into the mobile delivery application using their secure credentials. Once authenticated, the dashboard loads, showing a list of assigned deliveries sorted by delivery time and optimized route order. The partner selects the first delivery from the list, which displays detailed information including the customer's name, address, contact number, and delivery notes. At the warehouse, the partner scans the barcode on the assigned package using the device camera, confirming the pickup. The system updates the order status in real-time from "Ready for Pickup" to "Out for Delivery" and records the pickup timestamp.

Using the integrated navigation feature within the app, the partner follows the turn-by-turn GPS directions to the delivery address. The system monitors the route and provides real-time traffic updates. Upon arrival, the partner taps the "Arrived" button, which triggers an automatic SMS notification to the customer: "Your delivery partner has arrived." The partner meets the customer, hands over the package, and requests a signature on the touchscreen of the device. Once the digital signature is captured, the partner selects "Mark as Delivered." The system immediately updates the order status to "Delivered," logs the GPS coordinates and delivery time, and triggers an automated workflow: a delivery confirmation email and an SMS containing the tracking ID and a link to rate the service are sent to the customer. The delivery partner then proceeds to the next assigned delivery.

Alternative Scenario 1: Customer Requests Safe Drop-off

As the delivery partner prepares to deliver the package, they notice a delivery instruction left by the customer during checkout: "Leave at doorstep if no answer." Instead of requiring a signature, the partner selects the "Safe Drop-off" option within the app. The system prompts the partner to take a photograph of the package at the delivery location as proof of placement. After capturing the image, the partner confirms the drop-off. The system updates the order status to "Delivered – Left at Door," saves the photo in the order record, and sends an automated notification to the customer with the photo and delivery confirmation. No physical signature is required.

Alternative Scenario 2: Delivery to a Pick-up Point

The customer chooses to have their package delivered to a pick-up location instead of their home. This location might be a store or a secure drop-off point in their area. The delivery driver goes to this pick-up spot instead of the customer's address. When they arrive, they scan the package's barcode at the location's check-in system. The system checks the barcode and gives the driver access to a secure storage area. The driver places the package inside and confirms the drop-off. The delivery app connects to the pick-up point's system and receives a confirmation number. The driver enters this number into the app, and the system updates the order status to "Delivered to Pick-up Point." The customer then gets an automatic text and email with the pick-up address, storage box number, and a unique code. The customer can collect their package from that location within the next few days.

Error Scenario 1: Bank Payment Failure for Cash-on-Delivery Orders

The delivery partner arrives at the customer's location with a Cash-on-Delivery (COD) order. The customer attempts to pay using a debit card via the partner's mobile card reader. The delivery partner processes the payment, but the transaction is declined by the bank due to insufficient funds or an expired card. The system receives the decline notification from the payment gateway and displays an error message on the partner's device: "Payment declined. Please request an alternative payment method." The delivery partner informs the customer, who then pays in cash instead. The partner manually updates the payment method in the app to "Cash" and completes the delivery. If the customer cannot pay, the partner selects "Payment Failed" and returns the package to the warehouse.

Error Scenario 2: System Authentication Failure During Check-in

At the start of the shift, the delivery partner attempts to log into the platform but receives an error message: "Authentication failed. Unable to verify credentials." This failure occurs because the central authentication service is temporarily unavailable due to a server outage. As a result, the partner cannot access the delivery list, scan packages, or update delivery statuses. The system logs the authentication failure and notifies the technical support team. The delivery coordinator is alerted and provides the partner with a manual backup list of deliveries for the day. Deliveries are tracked using a paper-based system until the authentication service is restored.

2.4 Effort/Time estimation calculation.

Lead: Shahd Loai

Contributors: Hala Sabobeh: discussion, Yara Qaraqe: reviewing, Layan Salem: discussion, Nour Omari: discussion.

MODIFIED

UR	Developers Can Work Concurrently	Estimated Effort (average / developer)	Total Effort (pw)
UR1 – Inventory Browsing	1 devs	2 pw	1 × 2 = 2 pw
UR2 – Search & Filtering	1 dev	2 pw	1 × 2 = 2 pw
UR3 – Online Purchase	2 devs	3 pw	2 × 3 = 6 pw
UR4 – After-Sales Services	2 devs	1 pw	2 × 1 = 2 pw
UR5 – In-Store Purchase	2 devs	2 pw	2 × 2 = 4 pw
UR6 – Home Delivery	1 dev	2 pw	1 × 2 = 2 pw
UR7 – Reviews & Ratings	1 devs	1 pw	1 × 1 = 1 pw
UR8 – Business Information	1 dev	1 pw	1 × 1 = 1 pw
Total Effort (sum of all URs)	2 + 2 + 6 + 2 + 4 + 2 + 1 + 1 = 20 pw		
Average Number of Developers Needed	Sum of all developers working per UR/ number of URs: (1 + 1 + 2 + 2 + 2 + 1 + 1 + 1) / 8 = 1.38 ≈ 1 dev.		
Schedule Time	"We add 30% extra time for holidays, delays, …etc " 20 pw × 1.30 = 26 weeks		
Cost Estimation	Average weekly salary = $250 per developer Total schedule time = 26 weeks Cost: 250 × 26 = $6,500		
Profit Margin (10% – 30%)	❖ Minimum cost (10% profit):6500 × 1.10 = $7,150 ❖ Maximum cost (30% profit):6500 × 1.30 = $8,450		

2.5 ACTORS analysis and their description.

Lead: Layan Salem

Contributors: Hala Sabobeh: discussion, Shahd Loai: reviewing, Yara Qaraqe: discussion, Nour Omari: discussion.

Actor	Semantics/Description
Delivery Partner	This actor represents an authenticated staff member who logs into the delivery system, views assigned deliveries, updates order statuses, collects signatures, and handles exceptions such as missed delivery or address errors.
Manager	This actor is an administrative user with elevated privileges who can log in/out, access dashboards, view sales analytics, generate and export reports (inventory, sales, feedback, revenue), manage product inventory, and oversee system operations.
Website Guest	This actor represents someone who is not registered on the system that can browse products and search products, apply filters, and create a new account
Customer (Order Recipient)	This actor represents a registered user who can log in/out and perform all guest functions plus: manage shopping cart, save wish lists, complete checkout and payment, view order history, submit reviews and ratings, and manage account profile.
Warehouse Staff	This actor represents an employee who prepares orders for dispatch. They scan items into the system, update inventory, assign packages to delivery partners, and confirm pickup statuses.
Customer Support Agent	This actor represents an employee who can log in/out to view and respond to customer inquiries, assist with order issues, process returns and refunds, and escalate complex cases to managers.
Bank Payment System	This is an external system actor that validates credit card information and returns authorization status with confirmation codes.
Email Notification System	This external system actor that sends automated emails including order confirmations and shipping notifications.

2.6 USE-CASE diagram

Lead: Hala Sabobeh 1220322

Contributors: Sahd Loai: reviewing, Yara Qaraqe: discussion, Layan Salem: discussion, Nour Omari: discussion

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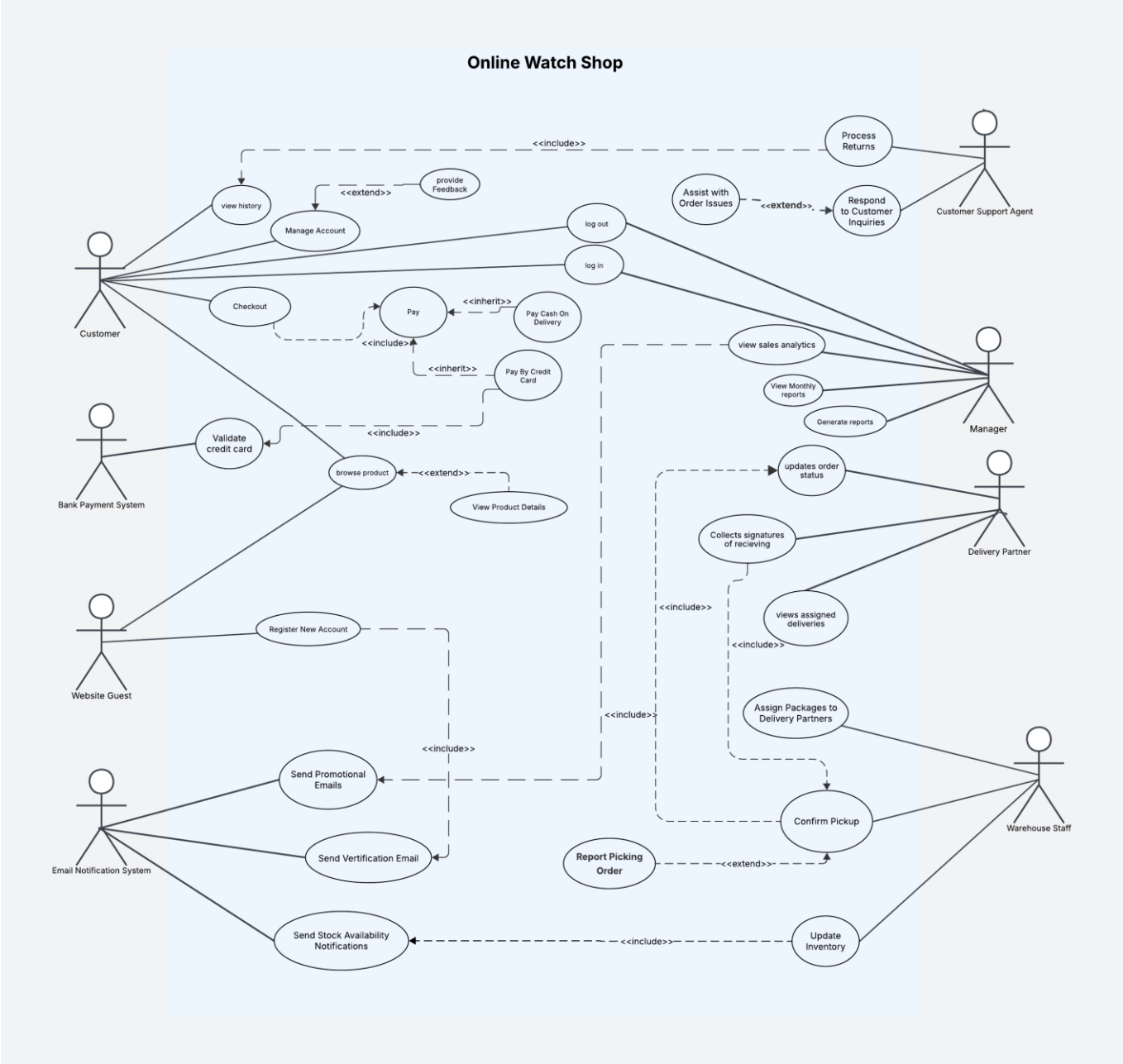


2.7 USE-CASE Specifications

Hala Sabobeh 1220322

Use Case Specification: Updates Order Status

Element	Description
Title	Updates Order Status
Description	The delivery partner updates the status of a delivery order at key stages: when picking up from warehouse, arriving at destination, and completing delivery. The system tracks each status change in real-time, records proof (signature/photo), and notifies the customer. If delivery cannot be completed (e.g., payment failure), the status is updated accordingly.
Actors	Delivery Partner (primary), Customer (secondary - receives notifications), Payment Gateway (for COD failure scenario)
Data	• Order ID and details • Current order status • GPS coordinates • Timestamps • Digital signature/photographic proof • Customer contact information • Package barcode • Payment information (for COD orders)
Trigger	The delivery partner initiates a delivery workflow by scanning a package barcode at the warehouse OR taps status buttons ("Arrived," "Mark as Delivered") during delivery.
Pre-conditions	1. Delivery partner is authenticated and logged into the mobile delivery app. 2. Partner has assigned deliveries in their queue. 3. Physical package is available with scannable barcode. 4. Mobile device has working GPS and data connection.
Normal Flow	1. Partner scans package barcode at warehouse using device camera. 2. System validates barcode against assigned deliveries. 3. **System updates status** to **"Out for Delivery"** and records pickup timestamp. 4. Partner navigates to delivery address using integrated GPS. 5. Upon arrival, partner taps "Arrived" button. 6. **System updates status** to **"Arrived at Location"**, records timestamp, and sends customer email. 7. Partner obtains recipient's digital signature on device screen. 8. Partner taps "Mark as Delivered." 9. **System updates status** to **"Delivered"**, stores signature/timestamp 10. System triggers automated customer notifications (email) with confirmation. 11. System displays next delivery in partner's queue.

Alternative Flows	**A. Safe Drop-off (No Signature):** 7a. Partner selects "Safe Drop-off" option (per customer instructions). 8a. System prompts partner to take photo of placed package. 9a. After photo capture, partner confirms. 10a. **System updates status** to **"Delivered – Left at Door"**, saves photo, and notifies customer with photo. **B. Delivery to Pick-up Point:** 5b. Partner navigates to designated pick-up location (not residential). 6b. Partner scans package at location's system or confirms in app. 7b. Partner receives/enters confirmation number from pick-up point system. 8b. **System updates status** to **"Delivered to Pick-up Point"** and sends customer pick-up instructions. **C. Customer Not Available (No Safe Drop Instructions):** 7c. Partner attempts delivery but no one is available to receive package. 8c. Partner selects "Customer Not Available" option. 9c. **System updates status** to **"Delivery Attempted – Customer Unavailable"** and schedules re-delivery.
Error Flows	**1. Invalid Barcode Scan:** • If scanned barcode doesn't match assigned deliveries, system displays "Invalid package. Please scan correct package." and does not update status. **2. COD Payment Failure:** • For Cash-on-Delivery orders, if electronic payment is declined and customer cannot pay: a. Partner selects "Payment Failed" option. b. **System updates status** to **"Delivery Attempt Failed - Payment Issue"**. c. System instructs partner to return package to warehouse. **3. GPS/Location Mismatch:** • If partner tries to update status but device GPS doesn't match delivery address coordinates, system displays "Cannot confirm location. Please ensure you're at the correct address." and prevents status update.
Post-conditions	1. Order status in system reflects final delivery outcome. 2. Complete audit trail exists (timestamps, GPS, proof of delivery). 3. Customer has received appropriate notification. 4. Delivery partner's queue advances to next assignment. 5. All data is synchronized with central database.
Comments	• Status updates must synchronize in real-time (<5 seconds) across all systems. • The system must handle offline scenarios: status updates should queue locally when connection is lost and sync when restored. • For security, status changes are non-reversible except by authorized supervisors. • Proof of delivery (signature/photo) must be stored securely and be non-editable.

Shahd loai - 1211019

Use-Case Specification: Request After-Sales Service

Title	Request After-Sales Service
Description	This use case describes how a customer requests an after-sales service for a previously purchased watch. The service may include repair, battery replacement, or product return. The system validates the order, warranty or return eligibility, stores the service request, and provides a confirmation message with a service request number.
Actors	Customer (Primary Actor) Store Staff (Secondary Actor – Alternative Flow)
Data	Order number Watch details Warranty status Return period information Service type Service description
Trigger	The customer selects the "After-Sales Service" option from the system.
Pre-conditions	The customer is logged into the system. The customer has purchased a watch from the store. The customer has a valid order number. The system is available and operational.
Normal Flow	1. Customer opens After-Sales Service page. 2. System displays service request form. 3. Customer enters order number. 4. System validates order number. 5. System retrieves watch details and warranty status. 6. Customer selects service type. 7. Customer enters description (optional). 8. Customer submits request. 9. System validates information. 10. System stores request. 11. System generates request number. 12. System displays confirmation message.
Alternative Flows	A1: In-Store Service Request – Staff enters request and system confirms. A2: Return Request – System checks return period and shows instructions if valid.
Error Flows	Invalid order number. Expired warranty or return period. Missing required information.
Post-conditions	Service request stored. Service request number generated. Customer receives confirmation.
Comments	Real-time validation of warranty and return period. Handles concurrent requests. Requests are traceable by request number.

Layan Salem - 1221026

Use case specification: Browse Products

Use case Title	Browse Products
Description	This use case allows a customer (guest or registered) to browse available watches, apply filters, and view detailed product information including price and availability. Allows a customer (guest or registered) to explore the watch catalog by browsing categories, applying filters (brand, price, material, etc.), sorting results, and viewing detailed product information, including specifications, images, pricing, and real-time availability.
Actors	Customer (Guest or Registered User)
Data	1. Product catalog 2. Product information 3. Product availability status 4. Product price
Stimulus/Trigger	The customer accesses the product catalog page.
Pre-conditions	1. Hand Watches Shop system is operational. 2. Product catalog database is accessible. 3. Customer's device has an active internet connection.
Workflow	• **Normal Flow (Successful Case)** 1. The customer accesses the catalog page. 2. System retrieves and displays a paginated list of available watches with images, names, prices, and stock indicators. 3. The customer optionally applies filters. 4. System queries the database and updates the displayed list in real-time. 5. Customer may sort results 6. Customer clicks on a specific watch product card. 7. System displays the detailed product page with images, description, specifications, customer reviews, price, stock status, and an "Add to Cart" button. **Alternative Flow A1 – No Products Match Filters:** 4a. If no products match the selected filters, the system displays: "No watches found matching your criteria. Try adjusting your filters." 4b. System suggests popular or related categories. **Alternative Flow A2 – Customer Saves Filter Preferences (Registered Users):** 5a. If the customer is logged in, they may choose to "Save These Filters" for future visits. 5b. System stores the filter set in the user's profile. **Error Flow E1 – Catalog Data Unavailable:** 2a. If the system cannot retrieve catalog data due to a database error, it displays: "We're having trouble loading the catalog. Please try again in a moment." 2b. System logs the error for administrative review.
Post conditions	1. The product catalog or selected product information is displayed correctly to the customer. 2. No system data is modified.
Comments	• This use case is a prerequisite for View Product Details and Manage Shopping Cart use cases. • Browsing does not require user authentication.

Use case specification: update inventory

Title	Update Inventory
Description	This use case describes how the system updates product inventory levels when warehouse staff process items for dispatch, return, or stock adjustment. The system ensures that product quantities, availability status, and related information are updated accurately and in real time.
Actors	Warehouse staff
Data	Product ID Product quantity in stock Product availability status Order details (if applicable)
Trigger	The warehouse staff scans items for dispatch, return, or stock adjustment, which requires updating the inventory records.
Pre-conditions	The warehouse staff is logged into the system. The product exists in the inventory database. The system is operational and connected to the inventory database.
Workflow	**Normal Flow** 1. The warehouse staff initiates the inventory update process. 2. The warehouse staff scans the product items using the system. 3. The system retrieves the current inventory data for the scanned products. 4. The system updates the product quantity based on the operation (dispatch, return, or adjustment). 5. The system updates the product availability status if required. 6. The system saves the updated inventory information in the database. 7. The system displays a confirmation message indicating that the inventory has been updated successfully. **Alternative Flow** A1. If the updated quantity reaches zero, the system marks the product as unavailable and may notify subscribed customers through the Email Notification System. **Error Flows** E1. If the inventory database cannot be accessed, the system displays an error message and no changes are saved. E2. If the scanned product is invalid or does not exist, the system displays an error message and cancels the update process.
Post-Conditions	The inventory records are updated correctly. The product availability status reflects the current stock quantity. No inventory changes are saved if an error occurs.
Comments	The system must maintain real-time synchronization of inventory data. Concurrent updates must be handled correctly to prevent data inconsistency. All inventory updates should be logged for auditing purpose.

Yara Qaraqe-1220505

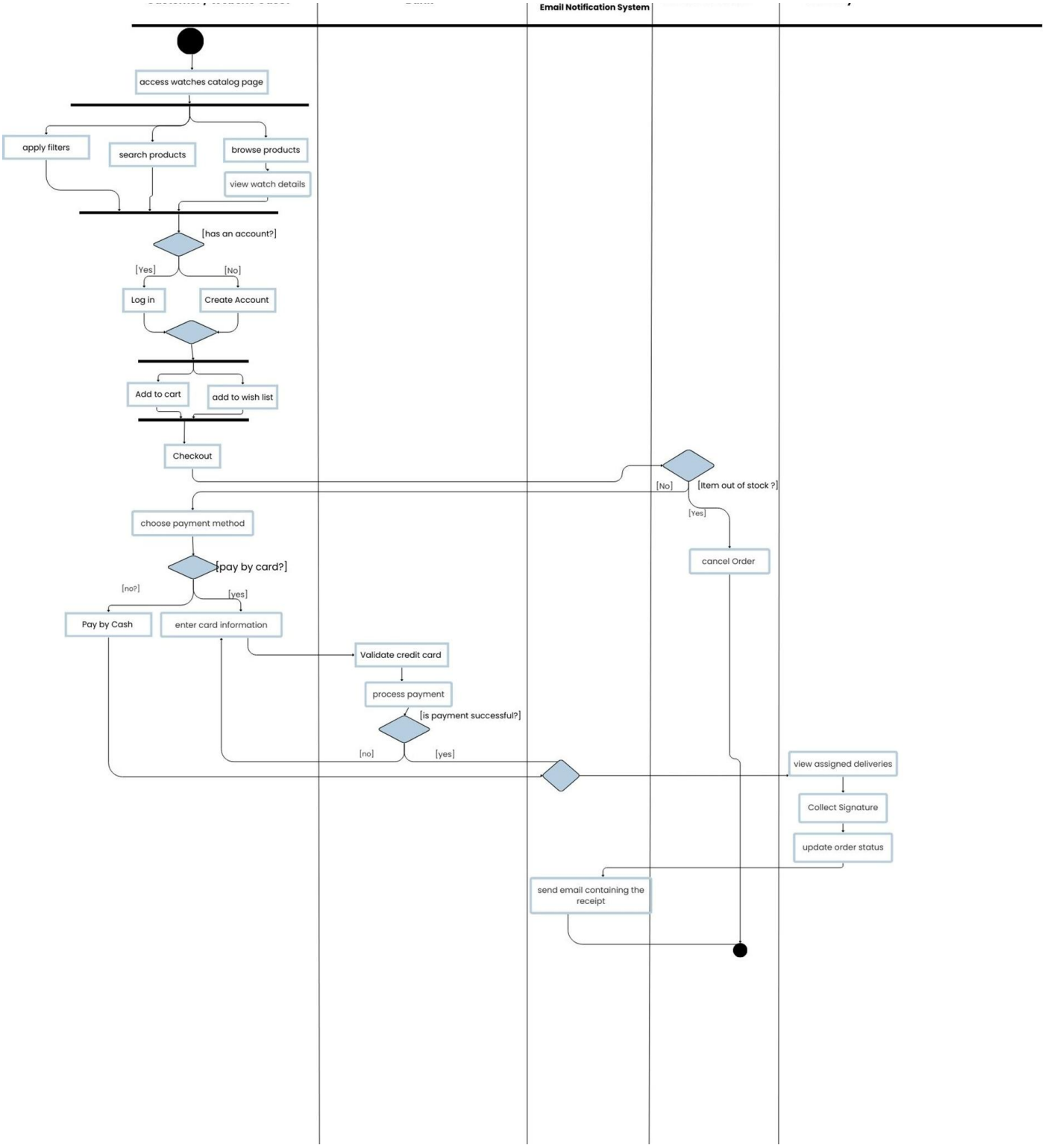
Use Case Specification: Complete Check out and Payment.

Use case Title	Complete Check out and Payment .
Description	A **Customer** may buy a watch from the store. The watch must exist in the warehouse and be available for purchase. The customer selects a watch and proceeds to checkout by clicking **"Proceed to Continue."** The system then displays a checkout request form, where the customer enters the required personal and delivery information and selects a payment method. The system displays a complete order summary for confirmation. Once the order is confirmed, the system generates a unique order number, reserves the ordered watch in the inventory, and sends an order confirmation email containing all order details to the customer.
Actors	Costumer (Primary actor), Bank Payment System (Secondary actor), Email Notification System (Secondary actor).
Data	Watch information, Customer information, Checkout information, Payment information, Order information, Inventory status information
Trigger	The costumer clicks the 'Proceed to Checkout' button from the shopping cart page.
Pre-conditions	1. The customer has at least one watch in the shopping cart. 2. Selected watches are available in inventory. 3. Payment Method is secured.
Workflow	**Normal Flow** 1. The customer reviews the shopping cart and verifies watch details. 2. The customer proceeds to checkout 3. The system displays the checkout form 4. The customer enters personal and delivery information. 5.The system validates the information and applies eligible discounts. 6. The customer selects home delivery and Cash on Delivery as the payment method. 7. The system displays the final order summary. 8. The system generates an order number, reserves the watch in inventory, and sends an order confirmation email. **Alternative Flow -Credit Card Payment** The customer selects **Credit Card** as the payment method. Then The system processes the payment successfully by directing the customer to a secure Payment Page, generates an order number, updates the inventory, and sends an email receipt **Error Scenario – Empty Shopping Cart** The customer attempts to proceed to checkout with an empty cart. Then The system prevents checkout and displays an error message.
Post-conditions	the system creates an order with a unique order number, updates the inventory, and displays and sends an order confirmation to the customer , If succeed no order is created and the inventory remains unchanged, If Failed
Comments	1. Order number is generated **only after successful payment or COD confirmation** **2. i**nventory updates must be atomic to avoid overselling. 3. Payment data must comply with security standards.

2.8 ACTIVITY diagram:

Lead: Nour Omari 1221212

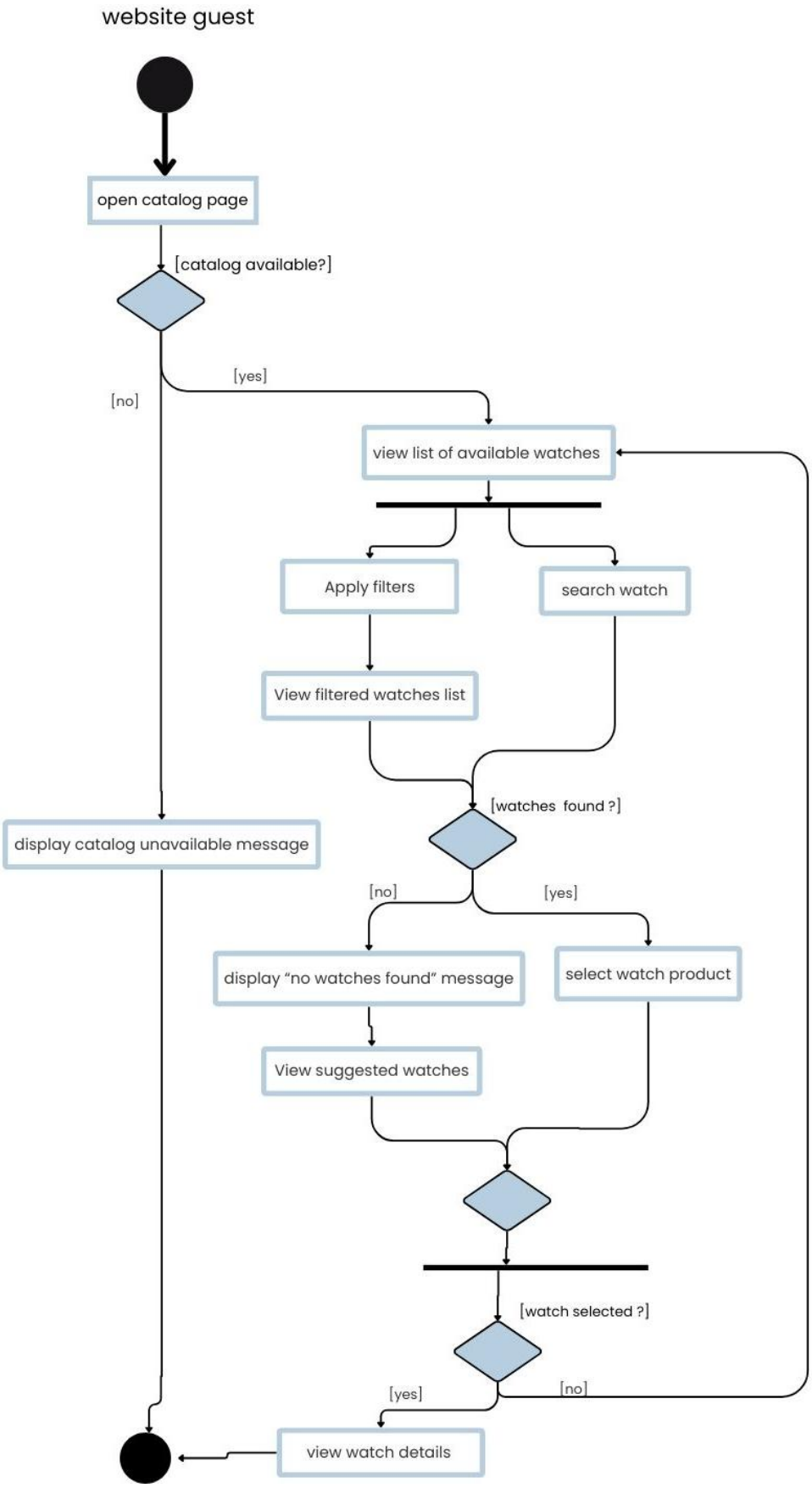
Contributors: Hala Sabobeh: reviewing, Layan Salem: reviewing, Shahd Loai: discussion, Yara Qaraqe :discussion



2.9 Instance Activity diagrams:

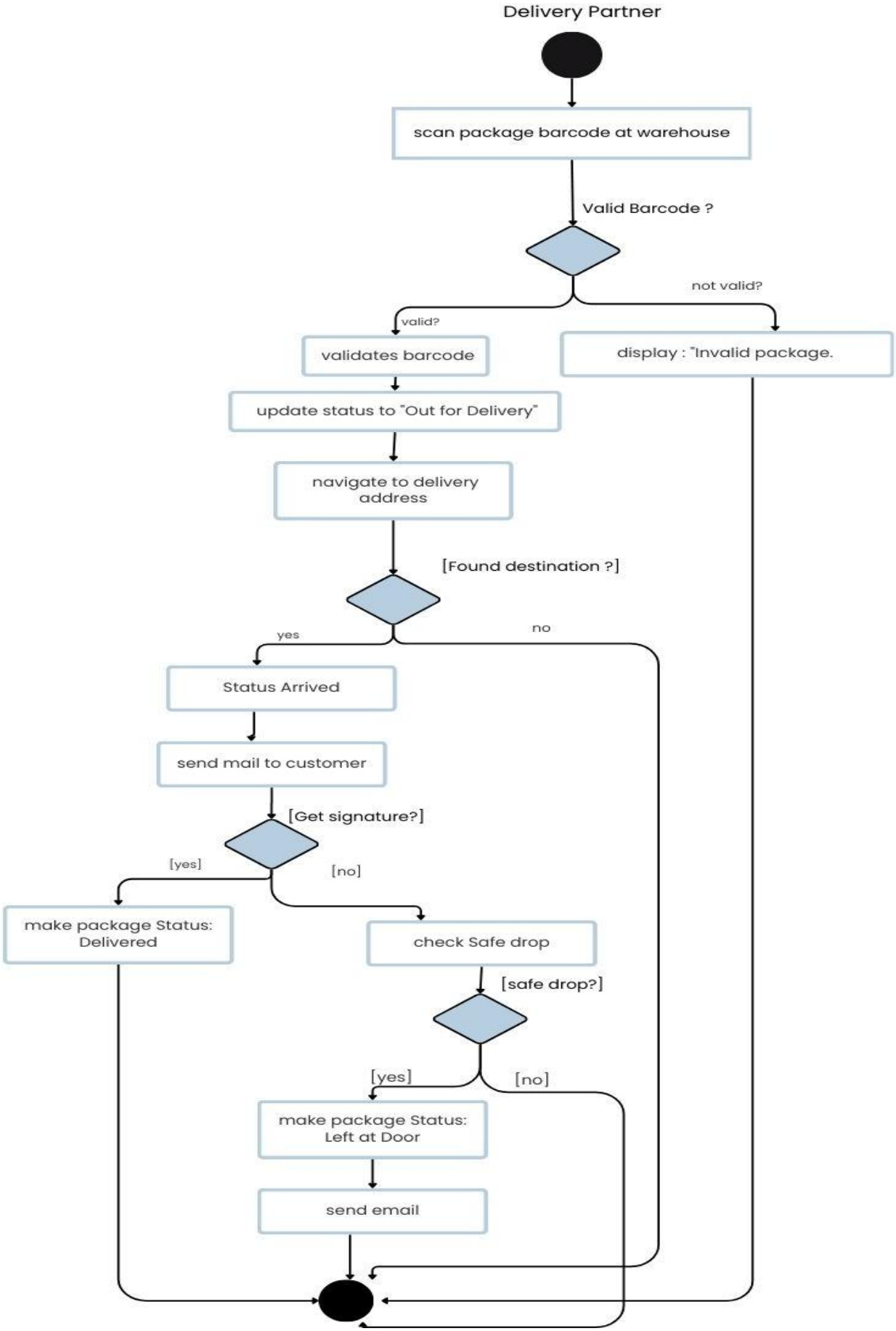
Layan Salem 1221026

Browse Products



Hala Sabobeh 1220322

update order status



Yara Qaraqe 1220505

Complete Check out and Payment .

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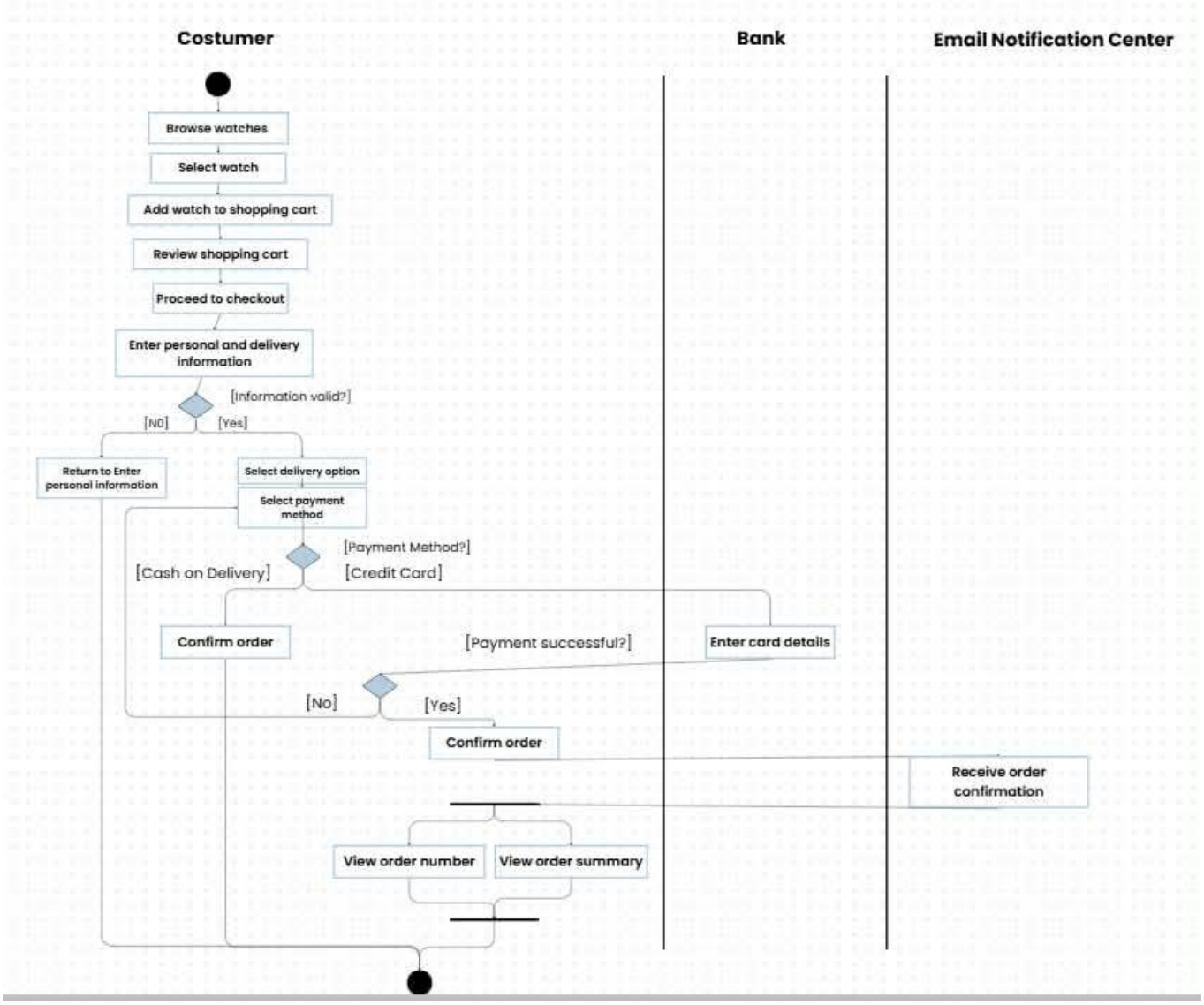
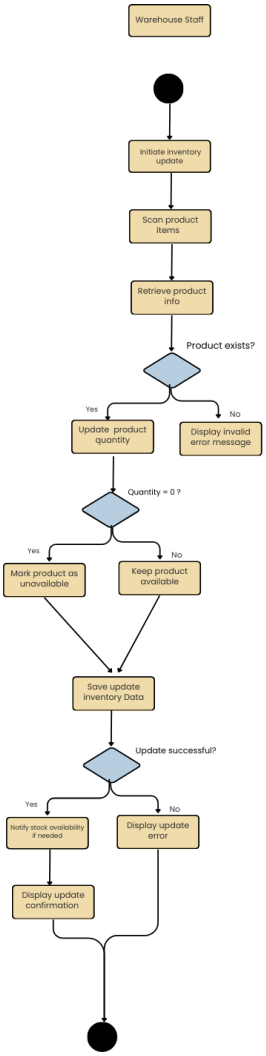
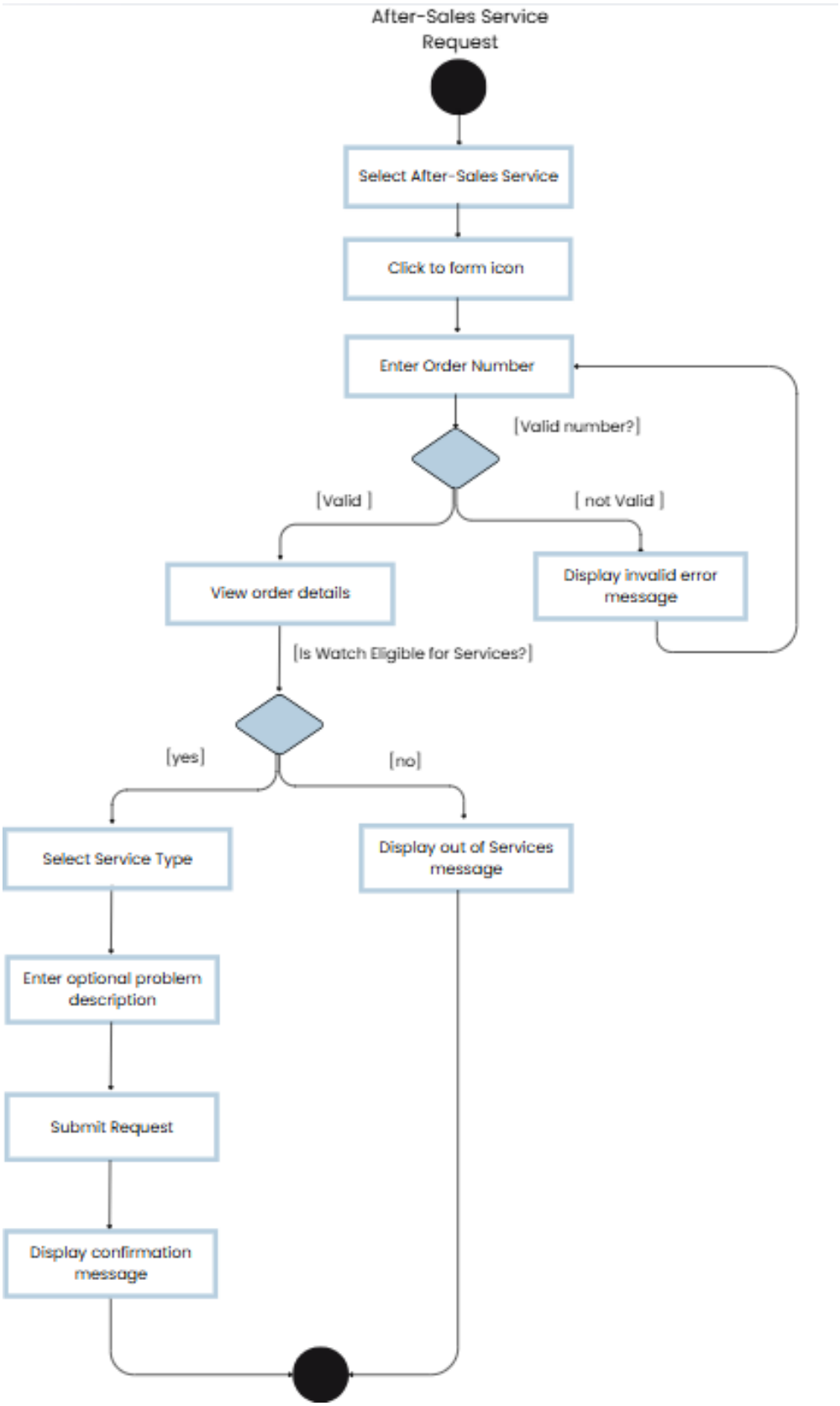


Nour Omarii 1221212

Update inventory




After-Sales Service Request
Select After-Sales Service
Click to form icon
Enter Order Number
[Valid number?]
[Valid]
[not Valid]
View order details
Display invalid error message
[Is Watch Eligible for Services?]
[yes]
[no]
Select Service Type
Display out of Services message
Enter optional problem description
Submit Request
Display confirmation message

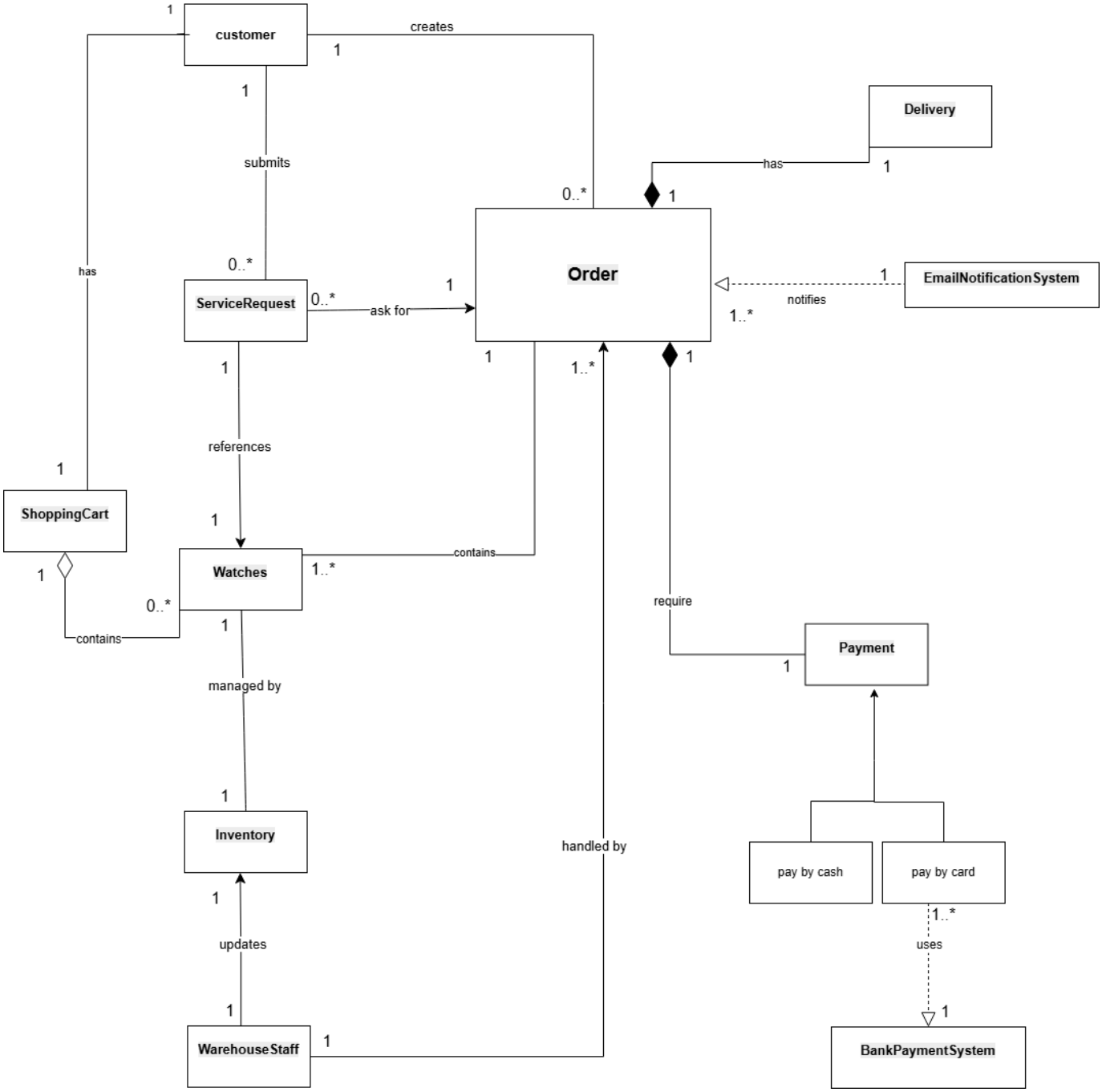
3 Chapter 3: System Analysis and Modelling

3.1 System CLASS Diagrams

Lead: Yara Qaraqe 1220505

Contributing members: Nour Omari (review), Shahd Loai (review), Layan Salem (discussion), Hala Sabobeh (discussion) MODIFIED

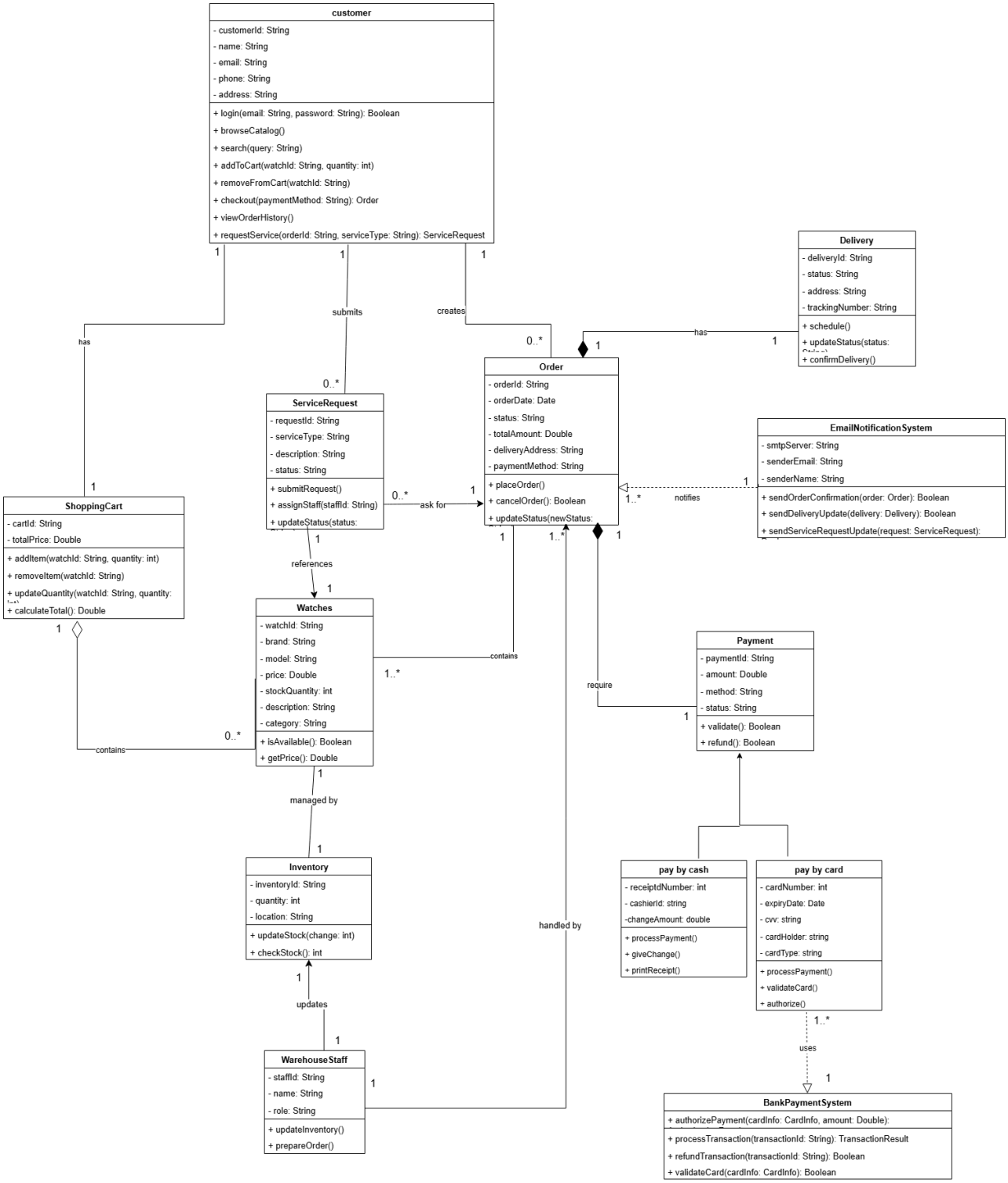
Analysis class model



Lead: Layan Salem 1221026

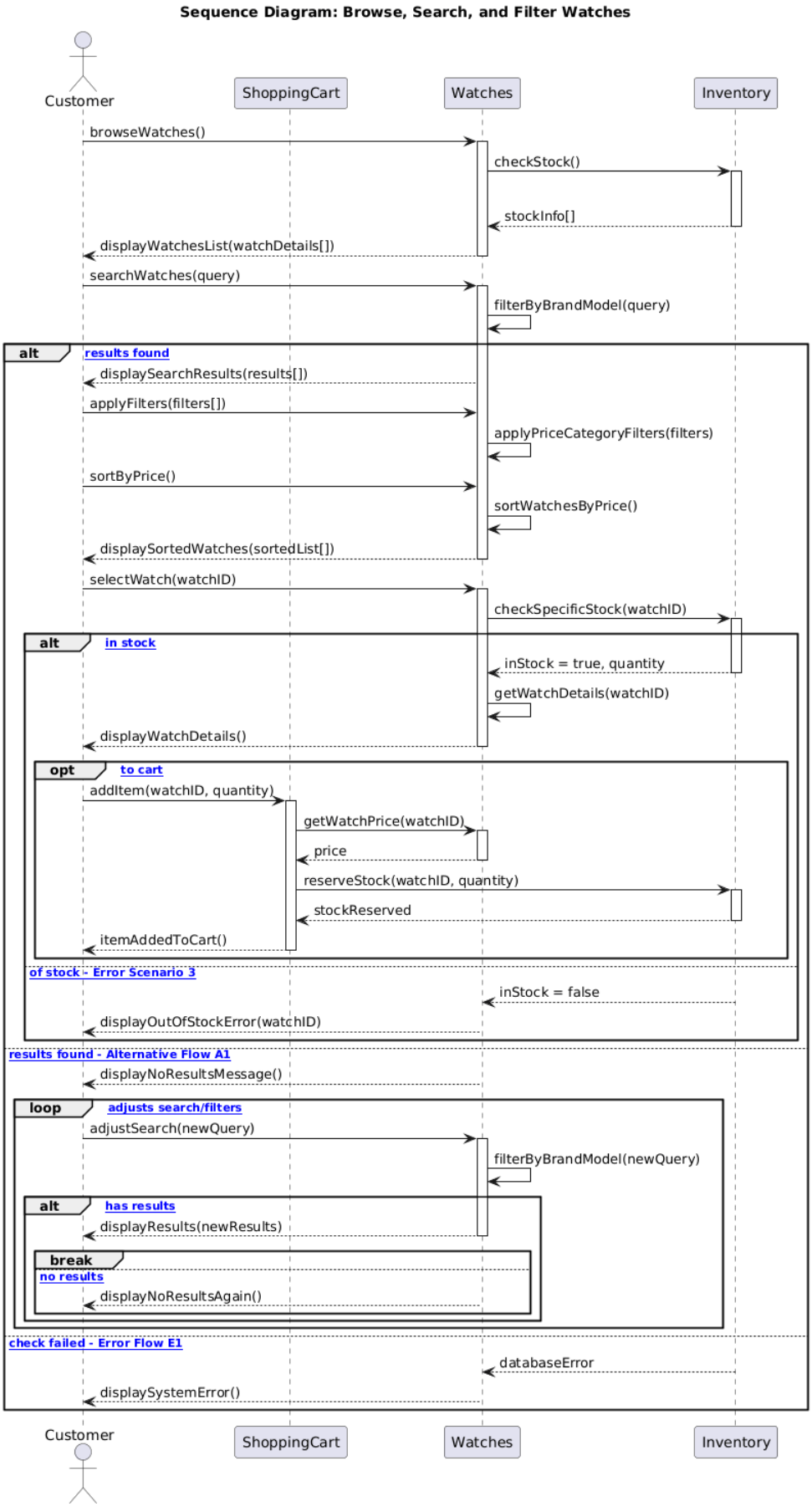
Contributing members: Nour Omari (review), Shahd Loai (review), Yara Qaraqe (discussion), Hala Sabobeh (discussion) MODIFIED

Detailed class model

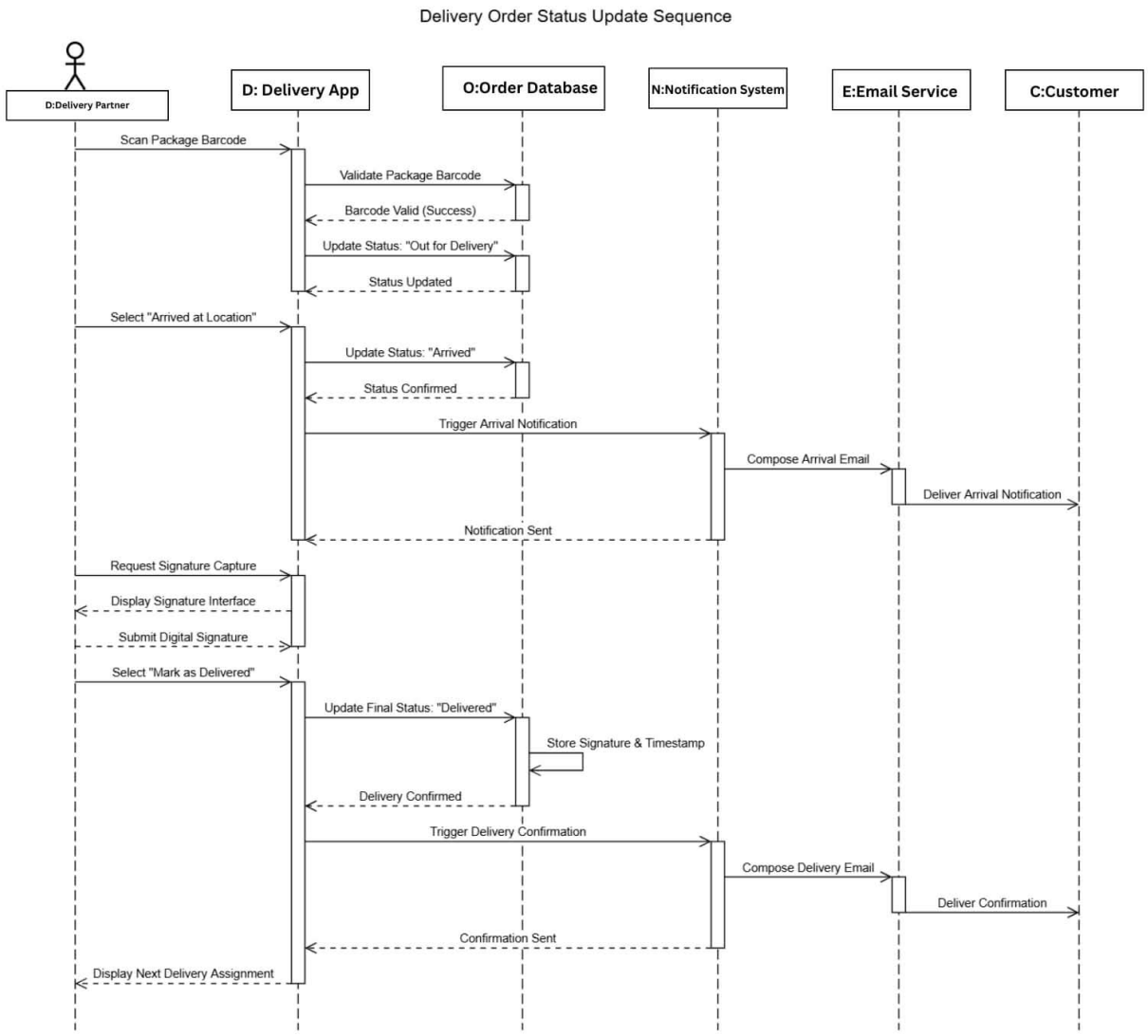


3.2 SEQUENCE Diagram

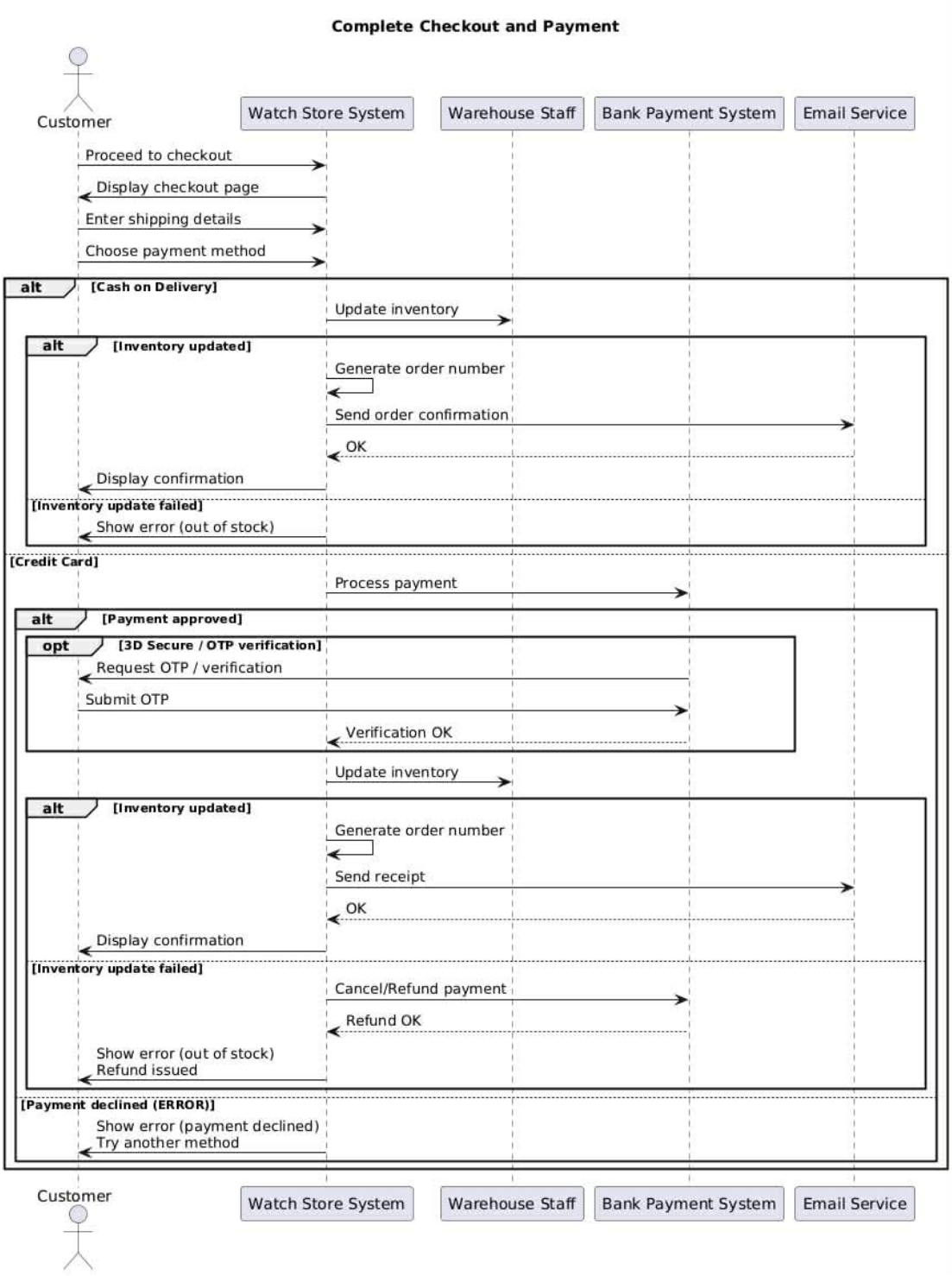
Layan Salem 1221026: Browse watches



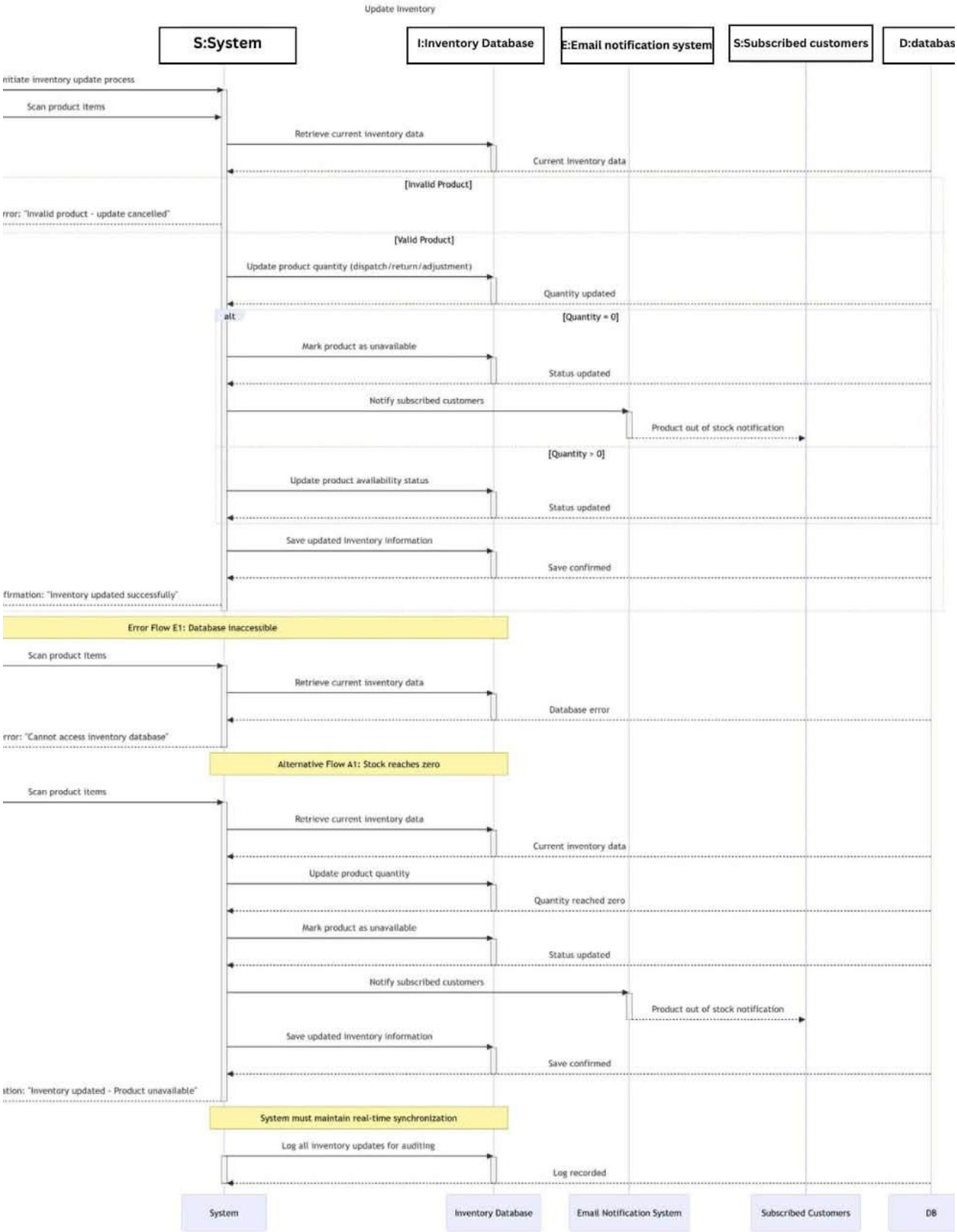
Hala Sabobeh 1220322: update order status



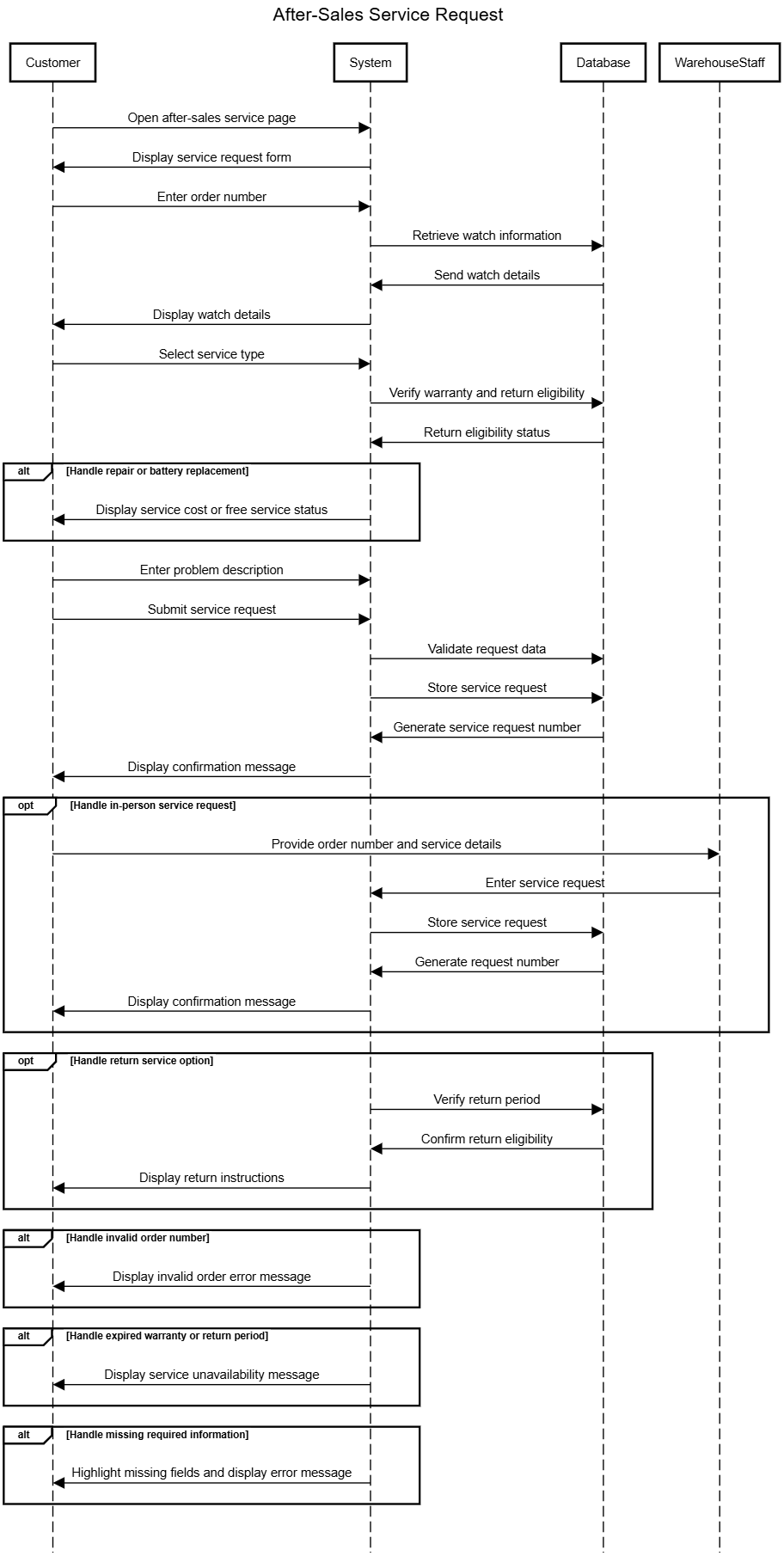
Yara Qaraqe 1220505: **Complete Check out and Payment**



Nour Omarii 1221212: Update inventory



Shahd Loai 1211019: after sales request



4 Chapter 4: System Design and Modelling

4.1 Description of chosen Design Goals

Lead: Nour Omari 1221212

Contributing members: Hala Sabobeh (review), Shahd Loai (review), Yara Qaraqe (discussion), Layan Salem (discussion)

The following design goals were selected to guide the design of the Watch Store Management System. These goals ensure good software quality at both class level and component level and align with software engineering best practices.

I. Low Coupling

Objective: Reduce dependencies between system classes and components to improve maintainability and flexibility.

Implementation:

At class level: Classes communicate through service interfaces using Dependency Injection. For example, OrderService interacts with PaymentService through an interface without accessing payment logic directly.

At component level: Core components (Checkout, Inventory, Delivery, After-Sales) are separated into independent services (Microservices). They communicate only through well-defined REST APIs.

Dependencies between components: Inventory Service depends on Order Service only for stock validation. Payment Service has no direct dependency on Inventory Service.

Deployment model: Each service is deployed independently using Docker containers, allowing updates without affecting other components.

II. High Cohesion

Objective: Ensure that each class and component has a single, well-defined responsibility.

Implementation:

At class level: Each class performs one function (e.g., InventoryService for stock updates, DeliveryStatusService for delivery tracking, PaymentProcessor for payment handling).

At component level: Each component handles one business domain:

Inventory Component → all inventory operations (stock updates, availability checks)

Order Component → all order processing (creation, tracking, cancellation)

Customer Component → all customer-related functions (profiles, history).

Mapping to use cases:

Update Inventory → Inventory Component

Complete Checkout → Order Component + Payment Component

Browse Products→ Catalog Component

How achieved: By mapping use cases directly to components and encapsulating internal logic within each component.

III. Specific System Design Goal

Derived from business non-functional requirements:

Performance: Inventory updates must propagate within < 2 seconds across all systems.

Reliability: high system availability; zero overselling of out-of-stock items.

Usability: Warehouse staff can learn the system in ≤ 1 day of training.

Specific Design Goal: System availability

Implementation:

Critical operations such as checkout and inventory updates are processed with real-time synchronization between Online Store, POS, Inventory, and Delivery components using WebSocket connections and Redis caching.

How achieved: Through transactional operations, real-time updates, and proper error handling for payment and inventory failures.

Business constraints: Must work on Android-based handheld scanners and sync with Windows POS systems; must handle offline scenarios when internet is unavailable.

Measurement criteria:

Processing speed: < 2000 ms for inventory updates

Fault tolerance: < 1 error per 1000 transactions

Training time: 1 day maximum for new staff

4.2 Component Diagram of your system components

Lead: Shahd Loai

Contributing members: Layan Salem(review), Yara Qaraqe (review), Nour Omari (discussion), Hala Sabobeh (discussion)

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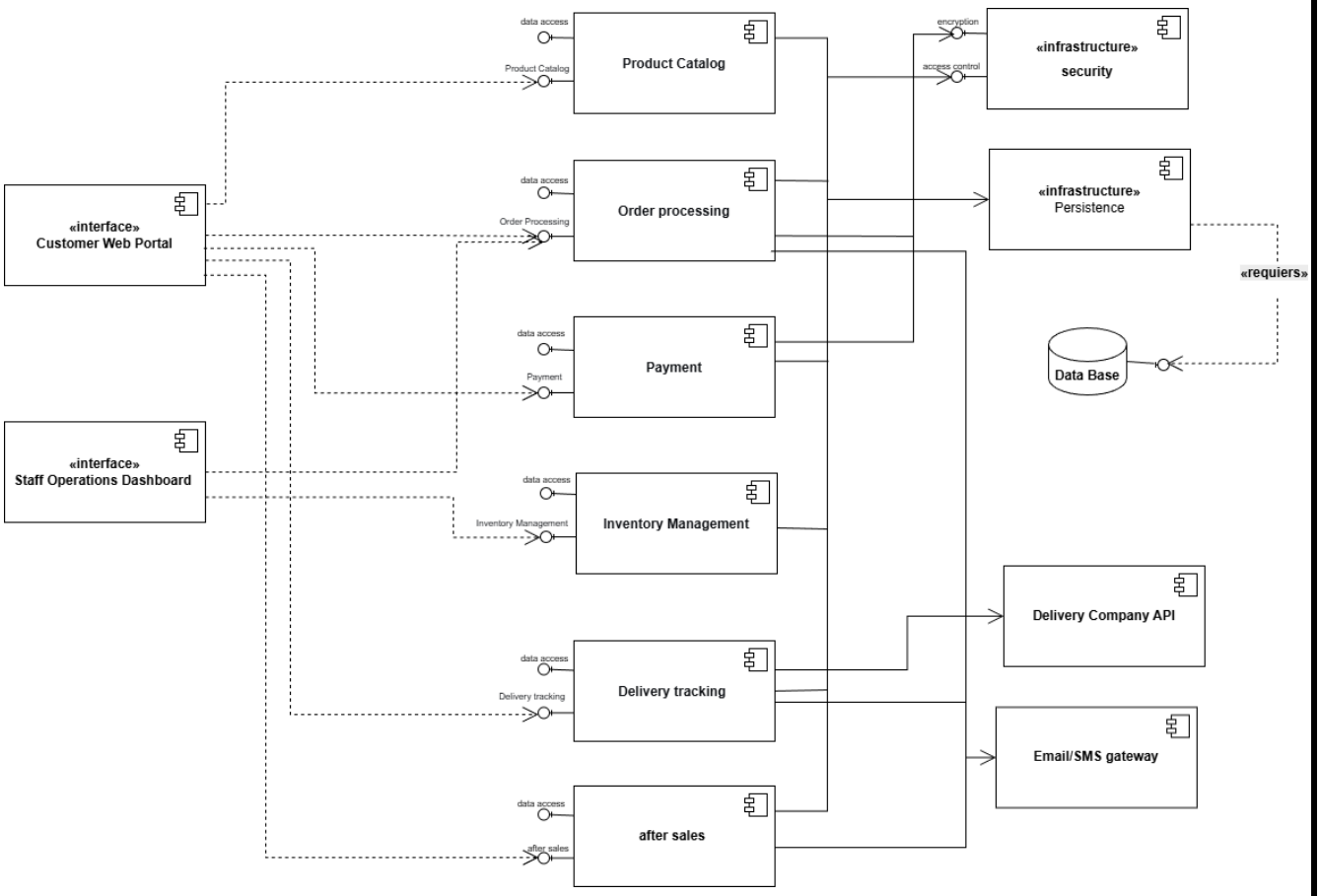

4.3 Overall architecture diagram

Lead: yara qaraqe

Contributing members: Layan Salem(review), Hala Sabobeh (review), Nour Omari (discussion), Shahd Loai (discussion)

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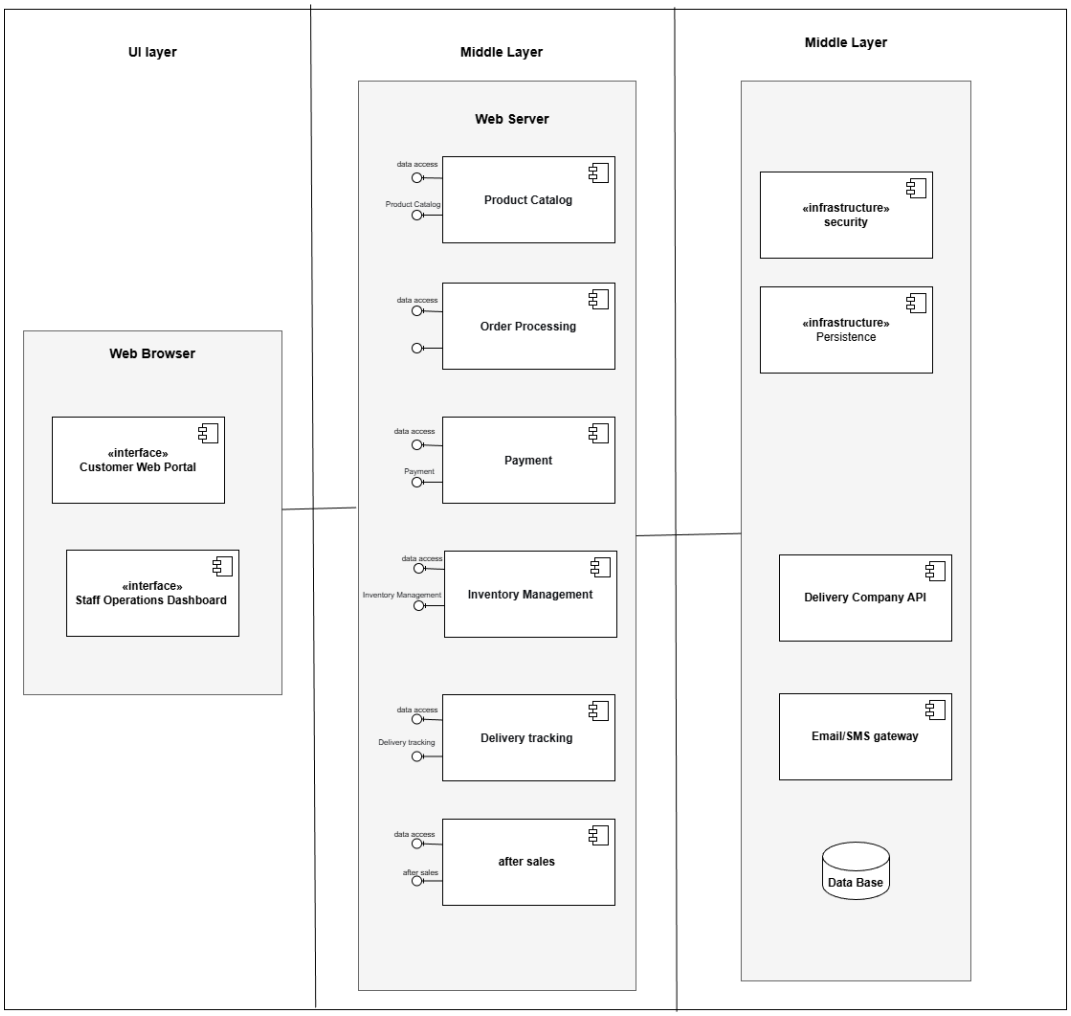

The chosen software architecture is a layered, component-based design that aligns with our system's key design goals and non-functional requirements. We adopted this approach because it clearly separates concerns such as presentation, business logic, and data access which supports high cohesion by ensuring each component has a single responsibility, like inventory management or order processing. It also enables low coupling by allowing independent services to communicate through well-defined APIs, making the system easier to maintain, scale, and update. This structure directly supports our performance and reliability goals, such as real-time inventory updates and fault tolerance, by using technologies like WebSockets and caching for synchronization. Additionally, the modular design accommodates business constraints like supporting both Android scanners and Windows POS systems, while allowing offline operation and future expansion without disrupting existing functionality.

4.4 Deployment diagram

Lead: Layan Salem

Contributing members: Yara Qaraqe(review), Hala Sabobeh (review), Nour Omari (discussion), Shahd Loai (discussion)

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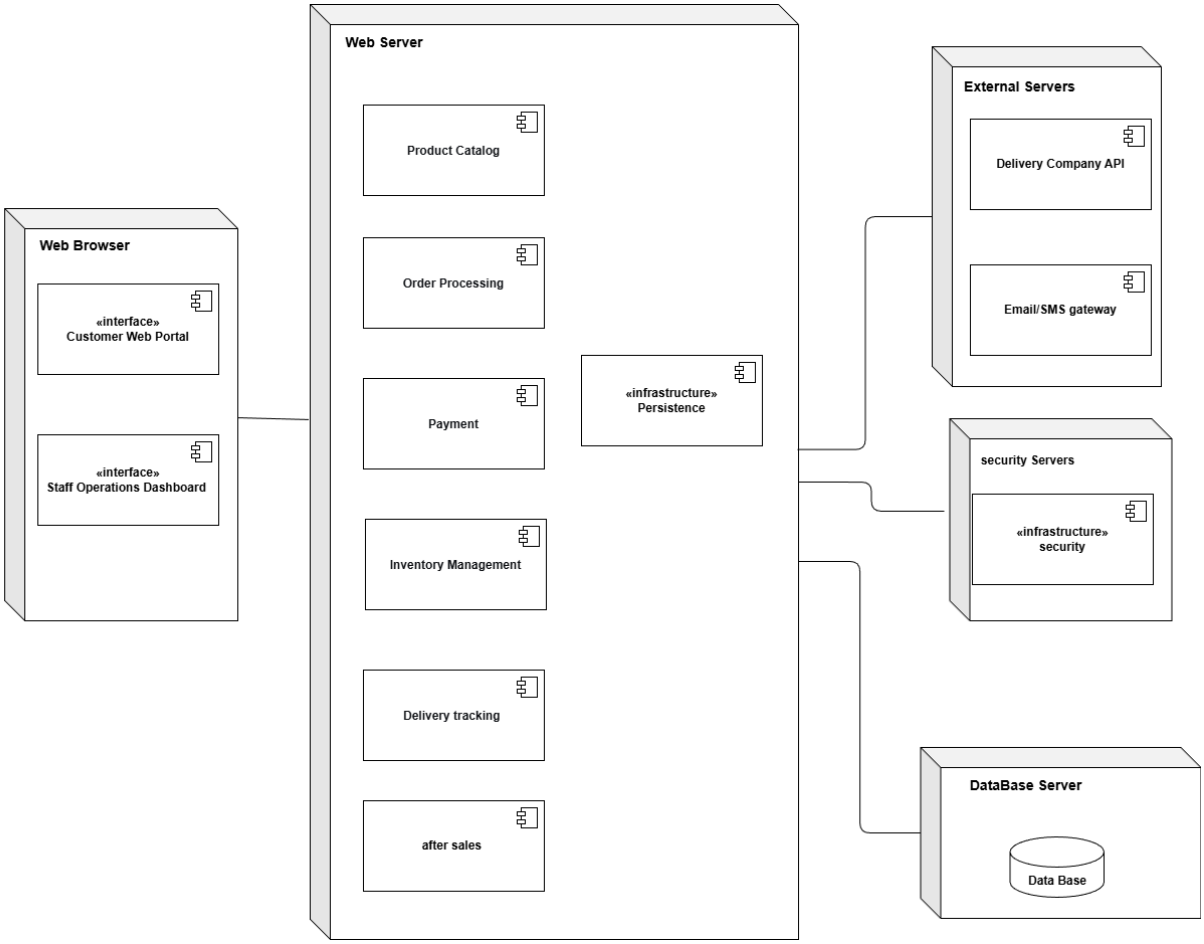

Appendix-I

Minutes of meetings

4.1 Table 1: Developer Group Internal Meetings

Meeting No.	Date	Time	Duration	Location	Attendees
2	4 Dec 2025	9:00 PM – 9:35 PM	35 mins	Google Meet	Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
3	7 Dec 2025	7:00 PM – 8:30 PM	1.5 hrs	Google Meet	Hala Sabobeh, Nour Omari, Shahd Loai
4	9 Dec 2025	8:00 PM – 10:00 PM	2 hrs	Google Meet	Yara Qaraqe, Layan Salem
5	13 Dec 2025	5:30 PM – 6:30 PM	1 hr	Google Meet	Yara Qaraqe, Layan Salem, Hala Sabobeh
6	25 Dec 2025	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
7	26 Dec 2025	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
8	27 Dec 2025	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
9	28 Dec 2025	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
10	29 Dec 2025	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
11	30 Dec 2025	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
12	31 Dec 2025	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
13	1 Jan 2026	5:00 PM – 8:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
14	3 Jan 2026	6:00 PM – 8:30 PM	2.5 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
15	6 Jan 2026	7:00 PM – 9:00 PM	2 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem
16	10 Jan 2026	5:30 PM – 8:00 PM	2.5 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
17	14 Jan 2026	6:00 PM – 8:00 PM	2 hrs	Google Meet	Yara Qaraqe, Layan Salem, Shahd Loai
18	18 Jan 2026	5:00 PM – 7:30 PM	2.5 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari
19	22 Jan 2026	7:00 PM – 10:00 PM	3 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
20	26 Jan 2026	3:00 PM – 5:00 PM	2 hrs	Face-to-Face	Yara Qaraqe, Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai
21	28 Jan 2026	6:30 PM – 9:00 PM	2.5 hrs	Google Meet	Yara Qaraqe, Hala Sabobeh, Layan Salem, Shahd Loai

4.2 Table 2: Meetings with Customer and Developer Group

Meeting No	Date	Time	Duration	Location	Developer Attendees	Customer Attendees
1	27 Nov 2025	During Class Session	20 mins	In-class	Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai, Yara Qaraqe	Omar Azam, Yara Khattab, Miassar Shamla (Johar), Taleen Bayatneh, Anas Shalabi
2	27 Nov 2025	During Class Session	20 mins	In-class	Haneen Nassar, Mohammed Muhaisen ,Alaa Nidal Abdelrahman Shaheen, Kyse Alqoran	Hala Sabobeh, Nour Omari, Shahd Loai, Yara Qaraqe, Layan Salem
3	4 Dec 2025	9:00 PM – 9:35 PM	35 mins	Google Meet	Hala Sabobeh, Layan Salem, Nour Omari, Shahd Loai	Omar Azam, Yara Khattab, Miassar Shamla (Johar), Taleen Bayatneh, Anas Shalabi
4	7 Dec 2025	7:00 PM – 8:30 PM	1.5 hours	Google Meet	Haneen Nassar, Mohammed Muhaisen ,Alaa Nidal Abdelrahman Shaheen, Kyse Alqoran	Hala Sabobeh, Nour Omari, Shahd Loai, Yara Qaraqe, Layan Salem